STAT 331

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Preface

These notes bring together material from *STAT 331 (Applied Linear Models)*, *STAT 444 (Statistical Learning – Advanced Regression)*, and the corresponding chapters of *The Elements of Statistical Learning*. They are written primarily for students and practitioners who wish to build a rigorous foundation in linear models, statistical learning theory, and their modern applications.

Although the formal coursework serves as the structural backbone of these notes, the perspective adopted throughout is influenced by the methods and expectations of quantitative research in finance. Systematic trading teams rely extensively on statistical modeling, and yet they often differ in the specific approaches they favour and in the technical depth they expect from candidates. Linear models, their generalizations, and their diagnostic tools remain central in many research pipelines and continue to be among the most frequently examined topics in interviews for fully systematic quantitative research roles.

The objective of this document is not only to reproduce the theoretical material, but also to present it in a way that reflects the type of mathematical clarity, interpretability, and modelling discipline demanded in quantitative finance. Derivations are shown explicitly, assumptions are stated precisely, and connections to practical model fitting are emphasized whenever relevant. The intention is to make these notes a useful companion both for academic study and for preparation for interviews in quantitative research, trading, and data-driven finance roles.

Readers are encouraged to approach the material actively: verify derivations, reproduce proofs, and apply the statistical principles to real data whenever possible. Mastery of linear models is not merely a mathematical exercise; it is the first step in understanding how systematic signals are built, validated, and ultimately deployed in production environments.

These notes will continue to expand as I progress through the courses and explore additional topics in statistical learning, regularization, dimension reduction, classification, and model selection.

1 Lecture 1: Overview of Linear Regression

Regression analysis involves modelling the *functional relationship* between *a response variable* y and *more than 1 explanatory variables* $x_1, \ldots, x_p$.

$$y = f(x_1, \ldots, x_p) + \epsilon$$

where ϵ is the random error term and y is our dependent variable. $f(\cdots)$ is a function that takes in *covariates/explanatory independent variables* or we will call *predictors*. The idea behind linear regression is to use linear models denoted by $y = f(x_1, \ldots, x_p) + \epsilon = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$ to model the relationship between y and the predictors $x_1, \ldots, x_p$. For the rest of these notes, we will let p be the number of features in our model and β be the *regression parameters*. Note that f does not depend on the parameters as we say that f *is linear in the parameters* β_j*'s*. Just to review, a *sample* is a collection of units actually measured/surveyed in a study. A *population* is a large group of units we are interested in, from which the sample is being selected.

> For purposes of these notes, we will assume that the data we have is representative of a larger population from which they are being sampled from.

For the sake of these notes, I will refrain from writing too much commentary as I believe that the proofs provided should suffice in justifying the results.

2 Lecture 2: Introduction to Simple Linear Regression

2.1 Simple Linear Regression

The **population model** consists of,

$$y = \underbrace{\beta_0 + \beta_1 x}_{\text{regression coefficients (systematic portion).}} + \underbrace{\varepsilon}_{\text{random error.}}$$

where β_0 is the *intercept* and β_1 is the *slope*. For sample (x_i, y_i) we can expand y_i as a function of x_i as $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. The following table summarizes the equation above.

	Fixed / Random	Known / Random
X_i	Fixed	Known
β_0, β_1	Fixed	Unknown
ε_i	Random	Unknown
y_i	Random	Known

2.2 Gauss-Markov Assumptions

(i) $\mathbb{E}[\varepsilon_i] = 0$.

(ii) ε_i's are statistically independent distributed errors.

(iii) $\text{Var}(\varepsilon_i) = \sigma^2$.

(iv) ε_i is normally distributed i.e., $\varepsilon_i \sim N(0, \sigma^2)$

Remark 2.1. Properties (i) and (iii) follow because $\mathbb{E}[y_i] = \beta_0 + \beta_1 x_i$ and $\text{Var}(y_i) = \sigma^2$.

The slope β_1 is often our primary interest,

(i) $\beta_1 = \mathbb{E}[y \mid x = a + 1] - \mathbb{E}[y \mid x = a]$.

(ii) If $\beta_1 = 0$ then $\mathbb{E}[y \mid x] = \beta_0$.

Remark 2.2. For now, we use lower case y and x to denote our response and explanatory variables as we are still dealing with univariate linear regression. We will reserve the utilisation of capital response and explanatory variables, Y and X respectively, for the latter half of these notes when we deal with multiple regression with vectors.

3 Lecture 3: Least Squares Estimation

Given samples N samples (x_i, y_i), we wish to find closed form equations for (β_0, β_1) such that the residual sum of squares is minimized. More explicitly, we wish to minimize,

$$\mathrm{RSS}(\beta_0, \beta_1) = \sum_{i=1}^{N} (y_i - \beta_0 - \beta_1 x_i)^2 .$$

We first differentiate $\dfrac{\partial \mathrm{RSS}(\beta_0, \beta_1)}{\partial \beta_0}$ and $\dfrac{\partial \mathrm{RSS}(\beta_0, \beta_1)}{\partial \beta_1}$ and set both derivatives equal to 0,

$$\begin{cases} \dfrac{\partial\, RSS(\beta_0, \beta_1)}{\partial \beta_0} = \displaystyle\sum_{i=1}^{N} 2\{\, y_i - \beta_0 - \beta_1 x_i\} \cdot (-1) = 0 \;\Rightarrow\; \sum_{i=1}^{N} (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (1) \\ \dfrac{\partial\, RSS(\beta_0, \beta_1)}{\partial \beta_1} = \displaystyle\sum_{i=1}^{N} 2\{\, y_i - \beta_0 - \beta_1 x_i\} \cdot (-1)\, x_i = 0 \;\Rightarrow\; \sum_{i=1}^{N} (y_i - \beta_0 - \beta_1 x_i) x_i = 0 \quad (2) \end{cases} .$$

We get the *least squares estimates* for $(\hat{\beta}_0, \hat{\beta}_1)$,

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\, \bar{x}, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

Before we continue with more derivations, I will make note of a couple useful formulas,

Concept	**Formula**
Sample variance	$s_x^2 = \dfrac{1}{n-1}\displaystyle\sum_{i=1}^{n}(x_i - \bar{x})^2 = \dfrac{1}{n-1} S_{xx}$
Sample covariance	$q_{xy} = \dfrac{1}{n-1}\displaystyle\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \dfrac{1}{n-1} S_{xy}$
Pearson correlation coefficient	$\rho_{xy} = \dfrac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$
Sample correlation r_{xy}	$r_{xy} = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$
OLS slope formulas	$\beta_1 = \dfrac{S_{xy}}{S_{xx}} = r_{xy}\,\dfrac{s_y}{s_x}$
Definition of S_{xx}	$S_{xx} = \displaystyle\sum_{i=1}^{n}(x_i - \bar{x})^2$
Definition of S_{xy}	$S_{xy} = \displaystyle\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$

Remark 3.1. S_{xx} and, equivalently, S_{yy} measure the spread of the predictor x around its mean and is proportional to the sample variance of x and y respectively.

Proof. Here we show that $\beta_0 = \bar{y} - \beta_1 \bar{x}$ and $\beta_1 = \dfrac{S_{xy}}{S_{xx}}$.

$$n\beta_0 = \sum_{i=1}^{N} y_i - \beta_1 \sum_{i=1}^{N} X_i \implies \beta_0 = \bar{y} - \beta_1 \bar{x},$$

$$0 = \sum_{i=1}^{N} x_i y_i - \beta_0 \sum_{i=1}^{N} x_i - \beta_1 \sum_{i=1}^{N} x_i^2 = \sum_{i=1}^{N} x_i y_i - \bar{y} \sum_{i=1}^{N} x_i + \beta_1 \bar{x} \sum_{i=1}^{N} x_i - \beta_1 \sum_{i=1}^{N} x_i^2,$$

Thus we have that,

$$\implies \beta_1 = \frac{\sum_{i=1}^{N} x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^{N} x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}.$$

Note that it is also possible to express β_1 as,

$$\beta_1 = \frac{q_{xy}}{S_x^2} = \rho_{xy}\frac{\sqrt{S_{yy}}}{\sqrt{S_{xx}}} \iff \beta_1 = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} = \rho_{xy}\frac{\sigma_y}{\sigma_x} = \frac{\sqrt{\frac{S_{xx}}{N-1}}}{\sqrt{\frac{S_{yy}}{N-1}}} = \rho_{xy}\frac{\sqrt{S_{yy}}}{\sqrt{S_{xx}}}.$$

i.e., the Pearson correlation coefficient scaled by the root ratio of sample variance of Y to the sample variance of X, denoted S_{yy} and S_{xx} respectively. □

3.1 Properties of $\hat{\beta}_0$ and $\hat{\beta}_1$.

(i) The *Least Squares Estimates* are *unbiased* i.e., $\mathbb{E}[\hat{\beta}_0] = \beta_0, \quad \mathbb{E}[\hat{\beta}_1] = \beta_1$.

(ii) $\mathrm{Var}(\hat{\beta}_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)$ and $\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$.

(iii) $\mathrm{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right) = -\frac{\sigma^2\bar{x}}{S_{xx}}$

Proof. We will prove the properties above. Let us define $c_i = -\frac{x_i - \bar{x}}{S_{xx}}$.

$$E[\hat{\beta}_1] = E\left(\sum_{i=1}^{N} c_i y_i\right) = \sum_{i=1}^{N} c_i E[y_i] = \sum_{i=1}^{N} c_i(\beta_0 + \beta_1 x_i) = \beta_0\sum_{i=1}^{N} c_i + \beta_1\sum_{i=1}^{N} c_i x_i = \beta_1$$

$$E[\hat{\beta}_0] = E[\bar{y} - \hat{\beta}_1\bar{x}] = E[\bar{y}] - \bar{x}\,E[\hat{\beta}_1] = E\left(\frac{1}{N}\sum_{i=1}^{N} y_i\right) - \bar{x}\,\beta_1 = \beta_0 + \beta_1\bar{x} - \beta_1\bar{x} = \beta_0$$

$$\mathrm{Var}(\hat{\beta}_1) = \mathrm{Var}\left(\sum_{i=1}^{N} c_i y_i\right) = \sum_{i=1}^{N} c_i^2\,\mathrm{Var}(y_i) = \sum_{i=1}^{N} c_i^2\sigma^2 = \sigma^2\sum_{i=1}^{N} c_i^2 = \frac{\sigma^2}{S_{xx}}$$

$$\begin{aligned}
\mathrm{Var}(\hat{\beta}_0) &= \mathrm{Var}(\bar{y} - \hat{\beta}_1\bar{x}) = \mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} y_i\right) + \bar{x}^2\ \mathrm{Var}(\hat{\beta}_1) \\
&= \mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} y_i\right) + \bar{x}^2\frac{\sigma^2}{S_{xx}} = \frac{1}{N^2}\sum_{i=1}^{N}\mathrm{Var}(y_i) + \bar{x}^2\frac{\sigma^2}{S_{xx}} \\
&= \frac{1}{N^2}(N\sigma^2) + \bar{x}^2\frac{\sigma^2}{S_{xx}} \\
&= \frac{\sigma^2}{N} + \frac{N\bar{x}^2}{S_{xx}}\,\sigma^2
\end{aligned}$$

$$\begin{aligned}
\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \mathrm{Cov}\left(\frac{1}{N}\sum_{i=1}^{N} y_i,\ \sum_{j=1}^{N} c_j y_j\right) = \sum_{i=1}^{N}\sum_{j=1}^{N}\frac{1}{N}c_j\,\mathrm{Cov}(y_i, y_j) \\
&= \sum_{i=1}^{N}\frac{1}{N}c_i\,\sigma^2 = \frac{\sigma^2}{N}\sum_{i=1}^{N} c_i \\
&= -\frac{\sigma^2\bar{x}}{S_{xx}}
\end{aligned}$$

□

Remark 3.2. It may appear at first that c_i was chosen arbitrarily however, it follows,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{1}{S_{xx}} \sum_{i=1}^N (x_i - \bar{x}) y_i - \frac{\bar{y}}{S_{xx}} \sum_{i=1}^N (x_i - \bar{x}) = \sum_{i=1}^N \frac{x_i - \bar{x}}{S_{xx}} y_i \implies c_i = \frac{x_i - \bar{x}}{S_{xx}}$$

Proposition 3.1. ***Properties of*** c_i

(i) ***Sum-to-Zero.***

$$\sum_{i=1}^N c_i = 0$$

(ii) ***Weighted Orthogonality.***

$$\begin{aligned}
\sum_{i=1}^N c_i x_i &= \frac{1}{S_{xx}} \sum_{i=1}^N x_i (x_i - \bar{x}) \\
&= \frac{1}{S_{xx}} \left(\sum_{i=1}^N x_i^2 - \bar{x} \sum_{i=1}^N x_i \right) = \frac{1}{S_{xx}} \left(\sum_{i=1}^N x_i^2 - N\bar{x}^2 \right) \\
&= \frac{1}{S_{xx}} \left(\sum_{i=1}^N (x_i - \bar{x})^2 \right) \\
&= \frac{S_{xx}}{S_{xx}} = 1
\end{aligned}$$

(iii) ***Squared-Weighted Identity.***

$$\sum_{i=1}^N c_i^2 = \sum_{i=1}^N \frac{(x_i - \bar{x})^2}{S_{xx}^2} = \frac{1}{S_{xx}^2} \sum_{i=1}^N (x_i - \bar{x})^2 = \frac{1}{S_{xx}}$$

Remark 3.3. Note that since we can write $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$, then we have that,

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \frac{\sum_{i=1}^N (x_i - \bar{x}) y_i}{S_{xx}} = \sum_{i=1}^N c_i y_i.$$

Note that c_i is a constant that only depends on x_i and not y_i.

3.2 Residuals

The ith residual is $r_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$.

Theorem 3.1. ***Properties of Residuals under LS.***

(i) $\sum_{i=1}^N r_i = 0.$

(ii) $\sum_{i=1}^N r_i x_i = 0 \implies \sum_{i=1}^N r_i \hat{y}_i = 0$ *from the fact* $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

Remark 3.4. By the the least squares estimates, we have two constraints on the residuals of our estimates. Therefore, if the ε_i are known then $\hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}\varepsilon_i^2$, or in other words,

$$\sigma^2 = \mathbb{E}[\varepsilon_i^2] = \mathbb{E}[\varepsilon_i]^2 + \mathrm{Var}(\varepsilon_i) = \sigma^2 + 0^2 = \sigma^2.$$

However, from the above, note that $\mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N} r_i^2\right] \neq \sigma^2$. By this result, if we define $S^2 = \frac{1}{N-2}\sum_{i=1}^{N} r_i^2$ then $\mathbb{E}[S^2] = \sigma^2$. Note here that $N-2$ is the degrees of freedom as we cannot count 2 of the values since we have 2 constraints on r_i.

Two of the r_i's cannot vary freely which is why sample variance for $y_i \sim N(0, \sigma^2)$ is defined as $\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{y})^2$ as a constraint on y_i is $\sum_{i=1}^{N}(y_i - \bar{y}) = 0$.

4 Lecture 4: Inference

For $\varepsilon_i \sim N(0, \sigma^2)$, we know from the theory of simple linear regression that $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$. This follows because $\hat{\beta}_1$ is a linear combination of the normally distributed errors. Recall from STAT 231 that if σ^2 were known, then

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim N(0,1),$$

so standard normal theory could be used directly. However, in practice we almost never know σ^2, so we replace it with the unbiased estimate S^2. This introduces additional uncertainty, and the resulting standardized quantity follows a t-distribution rather than a normal distribution,

$$\implies \frac{\hat{\beta}_1 - \beta_1}{\sqrt{S^2/S_{xx}}} \sim t_{n-2}.$$

Then, a $100(1-\alpha)\%$ confidence interval for β_1 is therefore,

$$\Pr\left(-t_{n-2,\,\alpha/2} < \frac{\hat{\beta}_1 - \beta_1}{\hat{\mathrm{SE}}(\hat{\beta}_1)} < t_{n-2,\,\alpha/2}\right) = 1-\alpha$$

$$\iff \quad \Pr\Big(\hat{\beta}_1 - t_{n-2,\,\alpha/2}\,\hat{\mathrm{SE}}(\hat{\beta}_1) < \beta_1 < \hat{\beta}_1 + t_{n-2,\,\alpha/2}\,\hat{\mathrm{SE}}(\hat{\beta}_1)\Big) = 1-\alpha$$

$$\iff \quad \left[\hat{\beta}_1 - t_{n-2,\,\alpha/2}\,\hat{\mathrm{SE}}(\hat{\beta}_1),\ \hat{\beta}_1 + t_{n-2,\,\alpha/2}\,\hat{\mathrm{SE}}(\hat{\beta}_1)\right].$$

Here $\mathrm{SE}(\hat{\beta}_1) = \sqrt{\dfrac{\sigma^2}{S_{xx}}}$ is the true (but unknown) standard error, and since σ^2 is unknown we use the estimate $\hat{\mathrm{SE}}(\hat{\beta}_1) = \sqrt{\dfrac{S^2}{S_{xx}}}$ instead. This substitution is precisely what leads to the use of the t-distribution with $n-2$ degrees of freedom in constructing the confidence interval.

4.1 Hypothesis Testing

We test $H_0 : \beta_1 = \beta_1^*$ against the alternative hypothesis $H_\alpha : \beta_1 \neq \beta^*$ under,

$$H_0 : t = \frac{\hat{\beta}_1 - \hat{\beta}_1^*}{\mathrm{SE}(\hat{\beta}_1)} \sim t_{n-2}.$$

We reject H_0 at the α significance level if $|t| \geq t_{n-2,\alpha/2}$. Alternatively we could use $p = P(|T| \geq |t|)$ for $T \sim t_{n-2}$ and reject if $p \leq a$. Usually, $H_0 : \beta_1 = 0$ is of interest.

4.2 Inference for Mean Response

For x_0 the mean response is $\mu_0 = \beta_0 + \beta_1 x_0$ and estimate $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.
Let the discrepancy measure be defined as,

$$d_i = \frac{1}{N}\frac{(x_0 - \bar{x})(x_i - \bar{x})}{S_{xx}},$$

and we wish to show that $\hat{\mu}_0 = \sum_{i=1}^{N} d_i y_i$.

Proof.

$$\begin{aligned}\hat{\mu}_0 &= \bar{y} - \hat{\beta}_1\bar{x} + \hat{\beta}_1 x_0 \\ &= \frac{1}{N}\sum_{i=1}^{N} y_i - \sum_{i=1}^{N} c_i y_i\,\bar{x} + \sum_{i=1}^{N} c_i y_i\, x_0 \\ &= \sum_{i=1}^{N}\left(\tfrac{1}{N} + (x_0 - \bar{x})c_i\right) y_i \\ &= \sum_{i=1}^{N} d_i y_i\end{aligned}$$

□

4.3 Distribution of μ_0

Theorem 4.1. ***Properties of*** $\hat{\mu}_0$

1. ***Unbiased.*** $\mathbb{E}[\hat{\mu}_0] = \mu_0$.

   Proof.

$$\begin{aligned}\mathbb{E}[\hat{\mu}_0] &= \mathbb{E}[\hat{\beta}_0 + \hat{\beta}_1 x_0] \\ &= \mathbb{E}[\hat{\beta}_0] + x_0\mathbb{E}[\hat{\beta}_1] \\ &= \beta_0 + \beta_1 x_0 \\ &= \mu_0.\end{aligned}$$

   □

2. $\mathrm{Var}(\hat{\mu}_0) = \left[\dfrac{1}{n} + \dfrac{(x_n - \bar{x})^2}{S_{xx}}\right]\sigma^2$.

   Proof.

$$\begin{aligned}\mathrm{Var}(\hat{\mu}_0) = \mathrm{Var}\left(\sum_{i=1}^{N} d_i y_i\right) &= \sum_{i=1}^{N} d_i^2\,\mathrm{Var}(y_i) \\ &= \left[\sum_{i=1}^{N}\frac{1}{N^2} + \sum_{i=1}^{N}\frac{2(x_0-\bar{x})(x_i-\bar{x})}{NS_{xx}} + \sum_{i=1}^{N}\frac{(x_0-\bar{x})^2 c_i^2}{S_{xx}}\right]\sigma^2 \\ &= \left[\frac{1}{N} + \sum_{i=1}^{N}\frac{(x_0-\bar{x})^2 c_i^2}{S_{xx}}\right]\sigma^2 \\ &= \left[\frac{1}{N} + \frac{(x_0-\bar{x})^2}{S_{xx}}\sum_{i=1}^{N} c_i^2\right]\sigma^2 \\ &= \left[\frac{1}{N} + \frac{(x_0-\bar{x})^2}{S_{xx}}\cdot\frac{1}{S_{xx}}\right]\sigma^2 \qquad \left(\text{property iii: } \sum_{i=1}^{N} c_i^2 = \tfrac{1}{S_{xx}}\right) \\ &= \left[\frac{1}{N} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]\sigma^2\end{aligned}$$

   □

4.4 Predicting Future Values.

We want the best guess of y given $x = x_p$, i.e., we want to guess y_p. Our best guess is

$$\hat{y}_p = \hat{\beta}_0 + \hat{\beta}_1 x_p.$$

(i) $\mathbb{E}[y_p - \hat{y}_p] = 0$ i.e., the predictor is unbiased.

Proof.

$$\begin{aligned}
\mathbb{E}[y_p - \hat{y}_p] &= \mathbb{E}[(\beta_0 + \beta_1 x_p + \varepsilon_p) - (\hat{\beta}_0 + \hat{\beta}_1 x_p)] \\
&= \mathbb{E}[(\beta_0 - \hat{\beta}_0) + (\beta_1 - \hat{\beta}_1)x_p + \varepsilon_p] \\
&= (\beta_0 - \mathbb{E}[\hat{\beta}_0]) + (\beta_1 - \mathbb{E}[\hat{\beta}_1])x_p + \mathbb{E}[\varepsilon_p] \\
&= (\beta_0 - \beta_0) + (\beta_1 - \beta_1)x_p + 0 \\
&= 0.
\end{aligned}$$

□

(ii) $\mathrm{Var}(y_p - \hat{y}_p) = \left[1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right]\sigma^2.$

Proof.

$$\begin{aligned}
y_p - \hat{y}_p &= (\beta_0 + \beta_1 x_p + \varepsilon_p) - (\hat{\beta}_0 + \hat{\beta}_1 x_p) \\
&= (\beta_0 - \hat{\beta}_0) + (\beta_1 - \hat{\beta}_1)x_p + \varepsilon_p \\
&= (\hat{\mu}_p - \mu_p) \; + \; \varepsilon_p,
\end{aligned}$$

where the prediction mean error satisfies

$$\mathrm{Var}(\hat{\mu}_p - \mu_p) = \left[\frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right]\sigma^2.$$

Since ε_p is independent of the fitted line and has variance σ^2, we obtain

$$\begin{aligned}
\mathrm{Var}(y_p - \hat{y}_p) &= \mathrm{Var}(\hat{\mu}_p - \mu_p) + \mathrm{Var}(\varepsilon_p) \\
&= \left[\frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right]\sigma^2 + \sigma^2 \\
&= \left[1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right]\sigma^2.
\end{aligned}$$

□

(iii) $\dfrac{y_p - \hat{y}_p}{\mathrm{SE}(y_p - \hat{y}_p)} \sim t_{n-2}.$

Proof.

$$\frac{y_p - \hat{y}_p}{\mathrm{SE}(y_p - \hat{y}_p)} = \frac{y_p - \hat{y}_p}{\sqrt{\left[1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right] S^2}} = \frac{(y_p - \hat{y}_p)/\sigma}{\sqrt{\left[1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}}\right] (S^2/\sigma^2)}},$$

where the numerator is normal with mean 0 and the denominator involves $\frac{(n-2)S^2}{\sigma^2} \sim \chi^2_{n-2}$.

Hence the ratio of a standard normal and the square root of an independent chi-square divided by its degrees of freedom yields

$$\frac{y_p - \hat{y}_p}{\mathrm{SE}(y_p - \hat{y}_p)} \sim t_{n-2}.$$

□

So the $100(1-\alpha)\%$ prediction interval for y_p is,

$$\hat{y}_p \pm t_{n-2,\alpha/2}\mathrm{SE}(y_p - \hat{y}_p).$$

Remark 4.1. We do not write $\mathbb{E}[\hat{y}_p] = y_p$ where y_p is also random so,

$$y_p - \hat{y}_p = \beta_0 + \beta_1 x_p + \varepsilon_p - (\hat{\beta}_0 + \hat{\beta}_1 x_p) = (\mu_p - \hat{\mu}_p) + \varepsilon_p.$$

So naturally, we get that,

$$\mathrm{Var}(y_p - \hat{y}_p) = \mathrm{Var}(\mu_p - \hat{\mu}_p) + \mathrm{Var}(\varepsilon_p) = \mathrm{Var}(\hat{\mu}_p) + \sigma^2.$$

5 Lecture 5: ANOVA and F-Statistic

5.1 Analysis of Variance of SST, SSR, SSE

Analysis of Variance for testing $H_0 : \beta_1 = 0$. We will use the following definitions:

Name	**Formula**	**Meaning**
Total Sum of Squares (SST)	$\text{SST} = \sum_{i=1}^{N}(y_i - \bar{y})^2$	total variation in y_i's
Sum of Squares of Regression (SSR)	$\text{SSR} = \sum_{i=1}^{N}(\hat{y}_i - \bar{y})^2$	variation explained by regression
Sum of Squares of Error (SSE)	$\text{SSE} = \sum_{i=1}^{N} r_i^2 = \sum_{i=1}^{N}(y_i - \hat{y}_i)^2$	unexplained variation

Table 1: Decomposition of Variation in Simple Linear Regression

Decomposition: $SST = SSR + SSE$. i.e., the Total Sum of Squares is equivalent to the sum of the Sum of Squares of Regression and Sum of Squares of Errors.

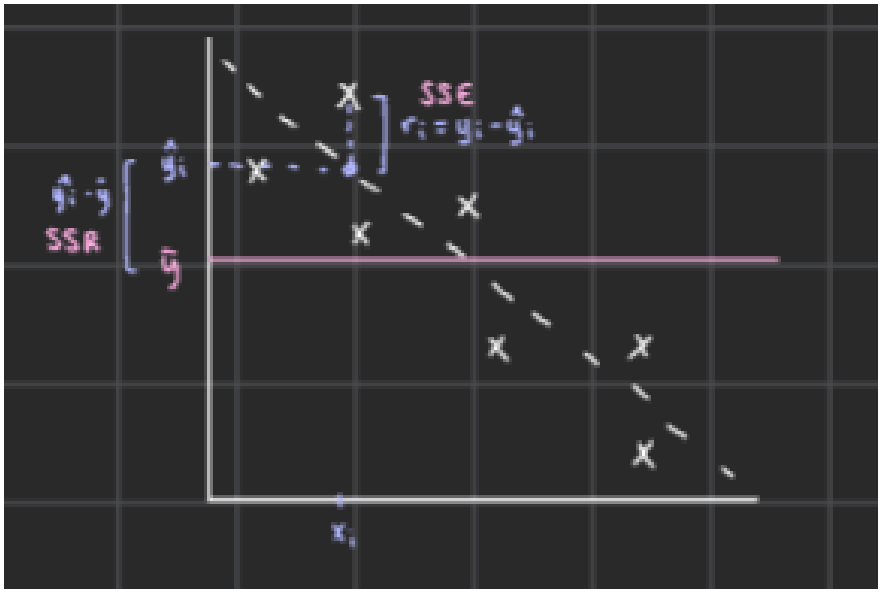


Figure 1: SSE vs. SSR Visualization.

Remark 5.1. In the diagram above, we observe that,

(i) The *Vertical Distance* between the datapoint y_i and the fitted line $\hat{y}_i$ is the error from the model, i.e., the variation the model fails to explain.

(ii) The *Horizontal Distance* between the datapoint y_i and the sample mean $\bar{y}$ is the variation that the regression line explains relative to the baseline predictor $\bar{y}$.

Proof.

$$\begin{aligned}
\text{SST} &= \sum_{i=1}^{N}(y_i - \bar{y})^2 \\
&= \sum_{i=1}^{N}(y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\
&= \sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \sum_{i=1}^{N}(\hat{y}_i - \bar{y})^2 + 2\sum_{i=1}^{N}(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \\
&= \text{SSE} + \text{SSR} + 2\sum_{i=1}^{N} r_i(\hat{y}_i - \bar{y}) \qquad \text{where } r_i = y_i - \hat{y}_i \\
&= \text{SSE} + \text{SSR} + 2\sum_{i=1}^{N} r_i(\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y}) \\
&= \text{SSE} + \text{SSR} + 2\hat{\beta}_1\sum_{i=1}^{N} r_i x_i + 2\hat{\beta}_0\sum_{i=1}^{N} r_i - 2\bar{y}\sum_{i=1}^{N} r_i \\
&= \text{SSE} + \text{SSR} + 2\hat{\beta}_1\sum_{i=1}^{N} r_i x_i + 2(\hat{\beta}_0 - \bar{y})\sum_{i=1}^{N} r_i \\
&= \text{SSE} + \text{SSR} \qquad \text{since } \sum_{i=1}^{N} r_i = 0 \text{ and } \sum_{i=1}^{N} r_i x_i = 0 \\
\Rightarrow \quad \text{SST} &= \text{SSR} + \text{SSE}.
\end{aligned}$$

□

Using this result we see that if $H_0 : \beta_1 = 0$ is true then $y_1 = \beta_0 + \epsilon_i$ then the Sum of Squares Regression remaining is small compared to the sum of squared errors as the regression line is approximately horizontal and close to $\hat{y}$. Under $H_0 : \beta_1 = 0$ we have the following results.

Theorem 5.1. ***Cochran's Theorem.***
Suppose $u_i \overset{iid}{\sim} N(0,1)$ *for* $i = 1, \ldots, n$ *and* $\sum_{i=1}^{n} u_i^2 = Q_1 + Q_2$. *Let* d_1 *and* d_2 *be the degrees of freedom of* Q_1 *and* Q_2, *i.e. # of linearly independent combinations of the* u_i*'s in* Q_1 *and* Q_2.
If $d_1 + d_2 = n$, Q_1 *and* Q_2 *are independent, and furthermore*

$$Q_1 \sim \chi^2_{d_1}, \qquad \textit{and} \qquad Q_2 \sim \chi^2_{d_2},$$

Theorem 5.2. ***Properties of SST, SSR, and SSE.***

(i) $SSR/\sigma^2 \sim \chi^2_1$

Proof.

$$\begin{aligned}
\text{SSR} &= \sum_{i=1}^{N}(\hat{y}_i - \bar{y})^2 = \sum_{i=1}^{N}(\beta_0 + \beta_1 x_i - \bar{y})^2 = \sum_{i=1}^{N}(\beta_1(x_i - \bar{x}))^2 \\
&= \hat{\beta}_1^2\sum_{i=1}^{N}(x_i - \bar{x})^2 = \hat{\beta}_1^2 S_{xx}
\end{aligned}$$

Recall that we previously had,

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right) \qquad \Longrightarrow \qquad \frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}} \sim N(0,1),$$

equivalent to the square of the $N(0,1)$. Thus under $H_0 : \beta_1 = 0$,

$$\frac{\hat{\beta}_1\sqrt{S_{xx}}}{\sigma} \sim N(0,1) \qquad \Longrightarrow \qquad \left(\frac{\hat{\beta}_1\sqrt{S_{xx}}}{\sigma}\right)^2 \sim \chi^2_1.$$

$$\frac{\text{SSR}}{\sigma^2} = \frac{\hat{\beta}_1^2 S_{xx}}{\sigma^2} = \left(\frac{\hat{\beta}_1\sqrt{S_{xx}}}{\sigma}\right)^2 \sim \chi^2_1.$$

□

(ii) $SST/\sigma^2 \sim \chi^2_{N-1}$

Proof. Under H_0, $\beta_1 = 0 \quad y_1, \ldots, y_n \overset{iid}{\sim} N(\beta_0, \sigma^2) \implies \dfrac{y_i - \beta_0}{\sigma} \sim N(0,1)$ thus $\displaystyle\sum_{i=1}^{n} \frac{(y_i - \beta_0)^2}{\sigma^2} \sim \chi^2_n$. Then, we have that,

$$\bar{y} \sim N\left(\beta_0, \frac{\sigma^2}{n}\right) \Rightarrow \frac{\bar{y} - \beta_0}{\sigma/\sqrt{n}} \sim N(0,1) \Rightarrow \left(\frac{\bar{y} - \beta_0}{\sigma/\sqrt{n}}\right)^2 \sim \chi^2_1.$$

$$\begin{aligned}
\text{SST} &= \sum_{i=1}^{N}(y_i - \bar{y})^2 = \sum_{i=1}^{N}\big((y_i - \beta_0) - (\bar{y} - \beta_0)\big)^2 \\
&= \sum_{i=1}^{N}(y_i - \beta_0)^2 \; - \; 2\sum_{i=1}^{N}(y_i - \beta_0)(\bar{y} - \beta_0) \; + \; \sum_{i=1}^{N}(\bar{y} - \beta_0)^2 \\
&= \sum_{i=1}^{N}(y_i - \beta_0)^2 \; - \; \underbrace{2(\bar{y} - \beta_0)\sum_{i=1}^{N}(y_i - \beta_0)}_{2N(\bar{y} - \beta_0)^2} \; + \; \underbrace{N(\bar{y} - \beta_0)^2}_{N(\bar{y} - \beta_0)^2} \\
&= \sum_{i=1}^{N}(y_i - \beta_0)^2 - N(\bar{y} - \beta_0)^2 \\
\Rightarrow \quad \sum_{i=1}^{N}\frac{(y_i - \beta_0)^2}{\sigma^2} &= \frac{\text{SST}}{\sigma^2} + \left(\frac{\bar{y} - \beta_0}{\sigma/\sqrt{N}}\right)^2 \implies \underbrace{\sum_{i=1}^{N}\frac{(y_i - \beta_0)^2}{\sigma^2}}_{S \sim \chi^2_N} = \underbrace{Q_1}_{\chi^2_N} + \underbrace{Q_2}_{\chi^2_1}
\end{aligned}$$

Therefore, we say that SST/σ^2 has $N-1$ degrees of freedom by definition of SST and given the constraint of $\sum_{i=1}^{N}(y_i - \bar{y}) = 0$. Since Q_1 and Q_2 are independent random variables, we can apply Cochran's Theorem to get the result. □

(iii) $SSE/\sigma^2 \sim \chi^2_{N-2}$ *and is independent of SSR.*

Proof. Recall that $SST = SSE + SSR$ so we can write $SST/\sigma^2 = SSE/\sigma^2 + SSR/\sigma^2$. Then $S = Q_1 + Q_2$. Given the result from (ii) we have that the degrees of freedom of the SSE is $N-2$. So since we know that Q_1 and Q_2 are independent, we know that $d_1 + d_2 = N - 1$, then we can apply Cochran's Theorem and get $SSE/\sigma^2 \sim \chi^2_{N-2}$. □

5.2 F-Statistic

We define the F-statistic as the following,

$$F = \underbrace{\frac{\frac{SSR}{\sigma^2}/1}{\frac{SSE}{\sigma^2}/N-2}}_{\text{expected value of } \chi^2_n \text{ is } n} = \frac{SSR}{SSE/N-2} = \underbrace{\frac{MSR}{MSE}}_{\text{regression over error}} \sim F(1, N-2).$$

i.e., $F(\text{df of numerator}, \text{df of denominator})$. To test $H_0 : \beta_1 = 0$ against $H_\alpha : \beta \neq 0$, we reject H_0 at α if $F > F_\alpha(1, N-2)$ which is the upper quantile (right tail) of the F distribution.

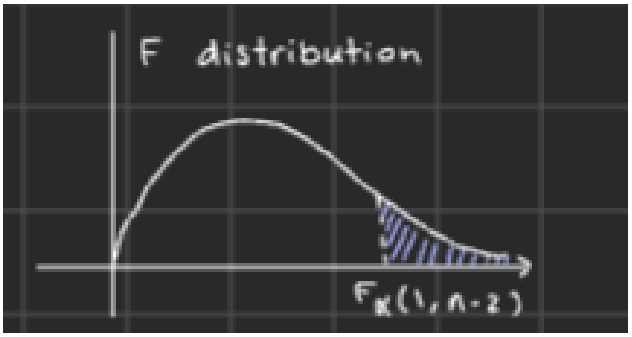


Figure 2: F distribution.

Recall that $\dfrac{\hat{\beta}_1 - \beta_1^*}{\text{SE}(\hat{\beta}_1)} \sim t_{N-2}$ can test $H_0 : \beta_1 = \beta_1^*$. In simple linear regression (univariate), testing $H_0 : \beta_1 = 0$ using the t-test is the same as using the F-test. Thus if $X \sim t_{N-2}$ we have that $X^2 \sim F(1, N-2)$.

Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Squares**	**F**
Regression	SSR	1	$\text{MSR} = \dfrac{\text{SSR}}{1}$	$\dfrac{\text{MSR}}{\text{MSE}}$
Residual	SSE	$N-2$	$\text{MSE} = \dfrac{\text{SSE}}{N-2}$	
Total	SST	$N-1$		

Theorem 5.3. ***Coefficient of Determination.*** *$R^2 = \frac{SSR}{SST}$. In simple linear regression, we have,*

$$R^2 = \frac{SSR}{SST} = \frac{\hat{\beta}_1^2 S_{xx}}{S_{yy}} = \left(\frac{S_{xy}}{S_{xx}}\right)^2 \frac{S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2$$

$$SSR = \sum_{i=1}^{N} (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^{N} (\beta_0 + \hat{\beta}_1 x_i - \bar{y})^2$$

$$\sum_{i=1}^{N} (\hat{\beta}_1 (x_i - \bar{x}))^2 = \hat{\beta}_1^2 \sum_{i=1}^{N} (x_i - \bar{x})^2 = \hat{\beta}_1^2 S_{xx}$$

6 Lecture 6: Linear Algebra and Multiple Linear Regression

6.1 Random Vectors and Matrices

Theorem 6.1. *Suppose that we have a vector* $Y = [y_1, \ldots, y_n]^\top$.
Then it is said to have the following properties:

(i) $\mathbb{E}[y_i] = \mu_i$ *and* $\mathrm{Var}(y_i) = \sigma_i^2$

(ii) $\mathrm{Cov}(y_i, y_j) = \sigma_{ij}$

(iii) $\mathrm{Var}(Y) = \mathbb{E}\{[Y - \mathbb{E}[Y]][Y - \mathbb{E}[Y]]^\top\}$

$$= \begin{pmatrix} \mathrm{Var}(y_1) & \mathrm{Cov}(y_1, y_2) & \mathrm{Cov}(y_1, y_3) & \cdots & \mathrm{Cov}(y_1, y_n) \\ \mathrm{Cov}(y_2, y_1) & \mathrm{Var}(y_2) & \mathrm{Cov}(y_2, y_3) & \cdots & \mathrm{Cov}(y_2, y_n) \\ \mathrm{Cov}(y_3, y_1) & \mathrm{Cov}(y_3, y_2) & \mathrm{Var}(y_3) & \cdots & \mathrm{Cov}(y_3, y_n) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(y_n, y_1) & \mathrm{Cov}(y_n, y_2) & \mathrm{Cov}(y_n, y_3) & \cdots & \mathrm{Var}(y_n) \end{pmatrix} = (\sigma_{ij})_{N \times N}.$$

Example 6.1. If $y_1, \ldots, y_n$ are iid then $\mathrm{Cov}(y_i, y_j) = 0$ for $i \neq j$ and $\mathrm{Var}(y_i) = \sigma^2$ for all i. Then, $\mathrm{Var}(Y) = \sigma^2 I$ where I is the $N \times N$ identity matrix.

Theorem 6.2. ***Expectation and Variance of Random Matrices.***

(i) $\mathbb{E}[AY + b] = A\mathbb{E}[Y] + b$ *for* $b \in \mathbb{R}^N$.

(ii) $\mathrm{Var}(Y + c) = \mathrm{Var}(Y)$ *(constant invariant)*

(iii) $\mathrm{Var}(AY) = A\mathrm{Var}(Y)A^\top$.

(iv) $\mathrm{Var}(AY + b) = A\mathrm{Var}(Y)A^\top$.

Proposition 6.1. *If we are differentiating a scalar function* f *wrt a vector* Y *then,*

(i) If $f = f(Y)$ *then* $\dfrac{df}{dY} = \left(\dfrac{\partial f}{\partial y_i}, \ldots, \dfrac{\partial f}{\partial y_n}\right)^2$.

(ii) If $f = c^\top Y = \sum_i c_i y_i$ *for some constant vector* c *then* $\frac{df}{dY} = c$.

(iii) If $A = A^\top$ *and* $f = Y^\top AY = \displaystyle\sum_{i=1}^{N}\sum_{j=1}^{N} a_{ij} y_i y_j$ *then* $\frac{df}{dY} = 2AY$.

Remark: *The double-sum in property (iii) is simply the quadratic form.*

Theorem 6.3. ***Matrix Results***

(i) ***Trace.*** *$tr(A) = \sum_i a_{ii}$ and $tr(BC) = tr(CB)$.*

(ii) ***Rank.*** *$rank(A)$ = maximum number of linearly independent columns (or rows).*

(iii) ***Independence.*** *$Y_1, \dots, Y_N$ are linearly independent iff the only solution to $c_1 y_1 + \cdots c_n y_n = 0$ is the trivial solution $c_i = 0$.*

(iv) ***Orthogonality.*** *X, Y are orthogonal matrices if $X^\top Y = 0$. Then A is an orthogonal matrix if $A^\top A = AA^\top = I$ i.e., $A^{-1} = A^\top$.*

Theorem 6.4. ***Eigenvalues and Idempotence.***

1. *v is an eigenvector of A if $Av = \lambda v$.*
2. ***Idempotent.*** *$A^2 = A$*
   (a) All eigenvalues are 0 or 1.
   (b) Idempotential + symmetric A implies that there exists an orthogonal matrix P such that $A = P\Lambda P^\top$ for $\Lambda = diag(1, \dots, 1, 0, \dots, 0)$.
   (c) $tr(A) = rank(A)$ = the number of eigenvalues that are equal to 1
   (d) $A^2 v = Av \implies \lambda^2 v = \lambda v \implies \lambda^2 = \lambda$.

Definition 6.1. Multivariate Normal Distribution. A random variable $Y \sim \text{MVN}(\mu, \Sigma)$ where Σ is the covariance matrix. The pdf of $Y = y$ is then given by,

$$f(y) = (2\pi)^{-N/2} |\Sigma|^{-1/2} \exp\left\{ -\frac{1}{2}(y-\mu)^\top \Sigma^{-1} (y-\mu) \right\}$$

where $\mathbb{E}[Y] = \mu$ and $\text{Var}(Y) = \Sigma$.

Theorem 6.5. ***Properties of MVN Random Variables.***

(i) ***Independence*** *iff Σ is a diagonal matrix i.e., $\text{Cov}(y_i, y_j) = 0$ iff y_i, y_j are independent*

(ii) ***Marginal Normality.*** *$y_i \sim N(\mu_i, \sigma_{ii})$*

(iii) If $Y \sim MVN(\mu, \Sigma)$ and $Z = AY$ then $Z \sim MVN(A\mu, A\Sigma A^\top)$.

(iv) If $Y \sim MVN(0, \sigma^2 I)$ then $\frac{1}{\sigma^2} Y^\top Y = \chi^2_n$.

6.2 Multiple Linear Regression

It is pretty unrealistic to assume that the univariate simple regression framework suffices. We move to a more realistic formulation where we have p fixed and known predictor variables of which there are $p+1$ fixed but unknown regression parameters. Then,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n$$

where now we have y_i's that are random and observable known responses with unknown random ε_i's. Similarly, we have assumptions consistent with simple linear regression.

Theorem 6.6. ***Multiple Regression Assumptions***

(i) $\mathbb{E}[\varepsilon_i] = 0$

(ii) $\mathrm{Var}(\varepsilon_i) = \sigma^2$.

(iii) $\varepsilon_1, \ldots, \varepsilon_N$ *are independent errors.*

(iv) $\varepsilon_1, \ldots, \varepsilon_N \sim N(0, \sigma^2)$ *iid for* $y_i \sim N(\beta_0 + \beta_1 x_i + \cdots \beta_p x_{i,p-1}, \sigma^2)$.

(v) If $U = AY$ *and* $W = BY$ *then* $\mathrm{Cov}(U, W) = A\Sigma B^\top$

Remark 6.1. Note that $U \perp W$ iff $A\Sigma B^\top = 0$.
i.e., U and W can be using different response variables y_i.

Proof.

$$\begin{aligned}
\mathrm{Cov}(U, W) &= \mathbb{E}\left\{[U - \mathbb{E}[U]][W - \mathbb{E}[W]]^\top\right\} \\
&= \mathbb{E}\left\{A[Y - \mathbb{E}[Y]][B(Y - \mathbb{E}[Y])]^\top\right\} \\
&= A\mathrm{Cov}(Y)B^\top \\
&= A\Sigma B^\top.
\end{aligned}$$

□

7 Lecture 7: Multiple Regression

7.1 Least Squares Estimation.

β_j quantifies the change in the response when the predictor X_j increases by one unit, while all other predictors are held constant. $H_0 : \beta_j = 0$ means X_j is not linearly related to the response y given all other predictors in the model.

Definition 7.1. Matrix Representation of Least Squares.
Let the feature matrix $X \in \mathbb{R}^{N\times(p+1)}$, the response vector $Y \in \mathbb{R}^N$, and the parameter vector $\beta \in \mathbb{R}^{(p+1)}$. Let the error vector be $\varepsilon \in \mathbb{R}^N$. Then the linear regression model can be written compactly as

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N1} & x_{N2} & \cdots & x_{Np} \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}}_{\beta} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{pmatrix}}_{\varepsilon}.$$

where each row of our feature matrix X represents the ith sample for $i = 0, 1, \ldots, N$ and each column of X represents the jth predictor. Each sample corresponds to a single y_i in our column vector of responses Y.

For a vector $\beta = (\beta_0, \ldots, \beta_p)^\top$ we aim to minimize the residual sum of squares.

$$\begin{aligned} \mathrm{RSS}(\beta) &= \sum_{i=1}^{N} (y_i - \beta_0 - \beta_1 x_{i,1} - \cdots - \beta_p x_{i,p})^2 \\ &= (Y - X\beta)^\top (Y - X\beta) \\ &= Y^\top Y - Y^\top X\beta - \beta^\top X^\top Y + \beta^\top X^\top X\beta \\ &= Y^\top Y - 2Y^\top X\beta + \beta^\top X^\top X\beta \end{aligned}$$

Taking the derivative given that $Y^\top X\beta = \beta^\top X^\top Y$ is a scalar,

$$\frac{dC^\top Y}{dY} = c, \quad \frac{dY^\top Ay}{dY} = 2AY.$$

and then taking the derivative of $\mathrm{RSS}(\beta)$ wrt the vector β and setting to 0 yields,

$$\begin{aligned} \frac{\mathrm{RSS}(\beta)}{d\beta} &= -2(X^\top Y) + 2X^\top X\beta = 0 \\ X^\top X\beta = X^\top Y &\implies \hat{\beta} = (X^\top X)^{-1} X^\top Y. \end{aligned}$$

Note that $X^\top X$ needs to be full rank (i.e. rank $p + 1$) in order for it to be invertible.

7.2 Properties of $\hat{\beta}$

Theorem 7.1. ***Properties of*** $\hat{\beta}$

(i) $\hat{\beta}$ is unbiased.

Proof.

$$\begin{aligned} \mathbb{E}[\hat{\beta}] &= \mathbb{E}[(X^\top X)^{-1} X^\top Y] \quad Y \sim \mathrm{MVN}(X\beta, \sigma^2 I), \\ &= (X^\top X)^{-1} X^\top \mathbb{E}[Y] \\ &= (X^\top X)^{-1} X^\top X\beta = \beta. \end{aligned}$$

□

(ii) $V(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$.

Proof.

$$\begin{aligned}\text{Var}(\hat{\beta}) &= \text{Var}[(X^\top X)^{-1}X^\top Y] \\ &= (X^\top X)^{-1}X^\top \text{Var}(Y)X[(X^\top X)^{-1}]^\top \\ &= (X^\top X)^{-1}X^\top \sigma^2 I X(X^\top Y)^{-1} \\ &= \sigma^2(X^\top X)^{-1}X^\top X(X^\top X)^{-1} \\ &= \sigma^2(X^\top X)^{-1}.\end{aligned}$$

Note that line two follows because $[(X^\top X)^{-1}]^\top = [(X^\top X)^\top]^{-1} = (X^\top X)^{-1}$.
Moreover, for square matrices, we know that $A^{-1}A = I$ implies that $(A^{-1}A)^\top = A^\top(A^{-1})^\top = I$ which means that $(A^\top)^{-1} = (A^{-1})^\top$. □

Remark 7.1. If $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ then $\hat{Y} = X\hat{\beta} = X(X^\top X)^{-1}X^\top Y$ are our fitted values. Suppose we let $H = X(X^\top X)^{-1}X^\top$ be our hat matrix. Then we have the following properties.

(i) H is idempotent i.e., $HH = H^2 = H$, and

$$HH = H^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = X(X^\top X)^{-1}X^\top = H.$$

(ii) H is symmetric i.e., $H = H^\top$.

8 Lecture 8: Important Results of Multiple Linear Regression

8.1 Review

Recall that MLR is of form $Y = X\beta + \varepsilon$ with the least squares estimate being $\hat{\beta} = (X^\top X)^{-1}X^\top \hat{Y}$ where $\hat{Y} = X\hat{\beta}$ (i.e., our fitted values). Naturally, $\mathbb{E}[\hat{\beta}] = \beta$ and $\text{Var}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$.

8.2 Residuals

(i) For $\hat{Y} = HY$ where $H = X(X^\top X)^{-1}X^\top$, H is idempotent and symmetric, i.e. $H = H^2$ and $H = H^\top$.

(ii) Let $r = Y - \hat{Y} = Y - HY = (I-H)Y$ be our residual vector. Then, $I - H$ is idempotent and symmetric,

$$\begin{cases} \sum_{i=1}^{N} r_i = 0, \\ X^\top r = 0, \\ \hat{Y}^\top r = 0 \end{cases} \quad \text{and} \quad \mathbb{E}[r] = 0, \quad \text{Var}(r) = \sigma^2(I-H).$$

(iii) The estimator $\hat{\sigma}^2 = \dfrac{\sum_{i=1}^{N} r_i^2}{N-(p+1)}$ is an *unbiased estimator* σ^2.

Proof. We start with our proof for the properties stated in (ii).

$$X^\top r = X^\top(Y - \hat{Y}) = X^\top Y - X^\top HY = X^\top Y - X^\top X(X^\top X)^{-1}X^\top Y = X^\top Y - X^\top Y = 0.$$

For $X^\top r$ we note that,

$$X^\top = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ X_{11} & X_{21} & \cdots & X_{n1} \\ \vdots & \vdots & \ddots & \vdots \\ X_{1p} & X_{2p} & \cdots & X_{np} \end{bmatrix}, \qquad X^\top r = \begin{bmatrix} \sum_i r_i \\ \vdots \\ \sum_i X_i r_i \end{bmatrix} = 0,$$

so we get that,

$$\hat{Y}^\top r = (\hat{\beta}^\top X^\top)\, r = \hat{\beta}^\top (X^\top r) \implies \hat{Y}^\top r = 0.$$

The expected value of our residuals is pretty straightforward,

$$\begin{aligned} E(r) &= E[(I-H)Y] = (I-H)E(Y) = (I-H)X\beta \\ &= X\beta - X(X^\top X)^{-1}X^\top X\beta \quad \underbrace{=}_{I} \quad 0. \end{aligned}$$

For variance, we note that generally, $\text{Var}(AY) = A\text{Var}(Y)A^\top$,

$$\begin{aligned} V(r) &= V((I-H)Y) \\ &= (I-H)\, V(Y)\, (I-H)^\top \qquad \text{(general identity)} \\ &= (I-H)\, \sigma^2 I\, (I-H)^\top \\ &= (I-H)\, \sigma^2 I\, (I-H) \qquad (I-H \text{ is symmetric}) \\ &= \sigma^2(I-H)(I-H) \qquad (I-H \text{ is idempotent}) \\ &= \sigma^2(I-H). \end{aligned}$$

For (iii), note that the number of β_j's corresponds to the number of degrees of freedom that we lose since we have $p+1$ constrains $X^\top_{(p+1)\times N} r = 0$. In simple linear regression we have $p = 1$ so $\hat{\sigma}^2 = \frac{\sum_{i=1}^N r_i^2}{N-2}$.

$$E\left(\sum_{i=1}^n r_i^2\right) = E(r^\top r) = E(\mathrm{tr}(rr^\top)) = \mathrm{tr}(E(rr^\top)) = \mathrm{tr}(V(r))$$

Note that $\mathrm{Var}(r) = \mathbb{E}[(r - \mathbb{E}[r])(r - \mathbb{E}[r])^\top]$ for $\mathbb{E}[r] = 0$. Moreover, in the trace, we can re-order with cylic permutations i.e. $\mathrm{tr}(ABC) = \mathrm{tr}(CAB) = \mathrm{tr}(BCA)$ thus,

$$\begin{aligned}
V(r) &= E\{[r - E(r)][r - E(r)]^\top\}, \qquad E(r) = 0 \\
\mathrm{tr}(V(r)) &= \mathrm{tr}(\sigma^2(I - H)) = \sigma^2[\mathrm{tr}(I) - \mathrm{tr}(H)] \\
&= \sigma^2[n - \mathrm{tr}(X(X^\top X)^{-1}X^\top)] \\
&= \sigma^2[n - \mathrm{tr}(X^\top X(X^\top X)^{-1})] \\
&= \sigma^2[n - \mathrm{tr}(I_{p+1})] \\
&= \sigma^2[n - (p+1)].
\end{aligned}$$

□

8.3 Sampling Distribution of β and $\hat{\sigma}^2$

Under normality, assume that $Y \sim \mathrm{MVN}(X\beta, \sigma^2 I)$ for $Y = X\beta + \varepsilon$.

Theorem 8.1. *Under normality, $\hat{\beta}$ has the following properties.*

(i) $\hat{\beta} \sim MVN(\beta, \sigma^2(X^\top X)^{-1})$.

(ii) $\hat{\beta}$ *and* σ^2 *are independent.*

(iii) $\dfrac{(N-p-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{N-p-1}$.

Proof.

(i) $\hat{\beta} = (X^\top X)^{-1}X^\top Y = (X^\top X)^{-1}X^\top(X\beta + \varepsilon) = \beta + (X^\top X)^{-1}X^\top\varepsilon$.

(ii) Recall that for any matrices A and B $\mathrm{Cov}(AY, BY) = A\,\mathrm{Var}(Y)\,B^\top$. We first observe that

$$\begin{aligned}
\hat{\beta} &= (X^\top X)^{-1}X^\top Y \sim MVN\big(\beta,\ \sigma^2(X^\top X)^{-1}\big)\,, \\
r &= (I - H)Y \sim MVN\big(0,\ \sigma^2(I - H)\big)\,,
\end{aligned}$$

so the least squares estimator and the residuals are (possibly degenerate) multivariate normal. Then,

$$\begin{aligned}
\mathrm{Cov}(\hat{\beta}, r) &= \mathrm{Cov}((X^\top X)^{-1}X^\top Y, (I - H)Y) = (X^\top X)^{-1}X^\top \mathrm{Var}(Y)(I - H)^\top \\
&= (X^\top X)^{-1}X^\top\sigma^2 I(I - H) \\
&= \sigma^2(X^\top X)^{-1}X^\top(I - X(X^\top X)^{-1}X^\top) \\
&= \sigma^2(X^\top X)^{-1}X^\top - \sigma^2(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top \\
&= \sigma^2(X^\top X)^{-1}X^\top - \sigma^2(X^\top X)^{-1}X^\top \\
&= \sigma^2(0) \\
&= 0 \in \mathbb{R}^{(p+1)\times N}.
\end{aligned}$$

So we have that $\hat{\beta} \perp r$ which means that $\hat{\sigma}^2 = \frac{r^\top r}{N-(p+1)}$ is a function of the residuals r.
Since $\hat{\beta} \perp r$ then $\hat{\beta} \perp r$ as well.

(iii) We let,

$$\hat{\sigma}^2 = \frac{r^\top r}{N-(p+1)} \implies \frac{(N-p-1)\hat{\sigma}^2}{\sigma^2} = \frac{r^\top r}{\sigma^2} = \left(\frac{r}{\sigma}\right)^\top \left(\frac{r}{\sigma}\right) = r^{*\top} r^*$$

where $r^* = \frac{r}{\sigma}$. Recall that our responses $Y \sim \text{MVN}(X\beta, \sigma^2 I)$ which means that,

$$r^* = \frac{(I-H)Y}{\sigma} \sim \text{MVN}(0, I-H).$$

We can use the eigendecomposition of symmetric an idempotent matrices i.e. there exists an orthogonal matrix P such that $P^\top P = PP^\top = I$. Note that $I - H = P\Lambda P^\top$ where,

$$\Lambda = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix},$$

and the number of ones on the diagonal equals the trace$(I-H) = N-p-1$. Define $Z \sim \text{MVN}(0, P^\top(I-H)P) \sim \text{MVN}(0, P^\top P\Lambda P^\top P) \sim \text{MVN}(0, \Lambda)$. Then, $z_1, \ldots, z_{N-p-1}$ are iid standard normal random variables. Moreover, $z_{N-p}, \ldots, z_N$ are all zeroes. Note that $z = P^\top r^* \iff PZ = PP^\top r^* = r^*$.

$$\frac{(N-p-1)\hat{\sigma}^2}{\sigma^2} = r^{*\top} r^* = (PZ)^\top (PZ) = Z^\top P^\top PZ = Z^\top Z \sim \chi^2_{N-p-1}.$$

□

9 Lecture 9: Inference under MLR

9.1 Inference of a Single β_j

(i) From (i) in **Theorem 8.1**, $\hat{\beta}_i \sim N(\beta_i, \sigma^2 v_{ii})$ (v_{ii} is ith diagonal element of $(X^\top X)^{-1}$)

(ii) From (ii) $\hat{\beta}$ and $\hat{\sigma}^2$ are independent.

(iii) From (iii) we saw $\dfrac{(N-p-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{N-p-1}$ so,

$$\implies \frac{\frac{\hat{\beta}_i - \beta_i}{\sigma^2 v_{ii}} \sim N(0,1)}{\sqrt{\frac{(N-p-1)\hat{\sigma}^2/\sigma^2}{N-p-1}} \sim \chi^2_{N-p-1}} = \frac{\hat{\beta}_i - \beta_i}{\sqrt{\underbrace{\sigma^2 v_{ii}}_{SE(\beta_i)}}} \sim t_{N-p-1}.$$

9.2 Inference on $\theta = a^\top \beta$

Suppose we have a scalar vector $a = [a_0, \ldots, a_p]^\top$ be a $(p+1)$-dim vector of constants. We would like to perform inference on $\theta = a^\top \beta = \sum_{i=0}^{p} a_i \beta_i$ i.e., estimate $\hat{\theta} = a^\top \hat{\beta}_i$. Since $\hat{\beta} \sim \text{MVN}(\beta, \sigma^2 (X^\top X)^{-1})$ then,

$$\hat{\theta} = a^\top \hat{\beta} \sim N(\theta = a^\top \beta, \sigma^2 a^\top (X^\top X)^{-1} a) \implies \frac{\hat{\theta} - \theta}{\sigma \sqrt{a^\top (X^\top X)^{-1} a}} \sim N(0,1) \quad \text{for known } \theta.$$

We usually do not know theta, so we use the estimator of sample variance,

$$\implies \frac{\hat{\theta} - \theta}{S \sqrt{a^\top (X^\top X)^{-1} a}} \sim t_{N-p-1}, \qquad \hat{\sigma} = \sqrt{\frac{SSE}{N-p-1}}.$$

9.3 Prediction Intervals

We want to predict y given predictor variables $X_1, X_2, \ldots, X_p$ and, $a_p = (1, x_1, \ldots, x_p)^\top y_p = a_p^\top \beta + \varepsilon_p$. We use the least squares estimates $\hat{\beta}$ for $\hat{y}_p = a_p^\top \hat{\beta}$. Then our $100(1-\alpha)\%$ prediction interval for y_p is,

$$\hat{y}_p \pm t_{N-p-1,\alpha/2} \text{SE}(y_p - \hat{y}_p).$$

Proof. For fixed $a_p^\top \beta$ and given the fact that $a^\top \hat{\beta} \sim N(\beta, \sigma^2 a^\top (X^\top X)^{-1} a)$,

$$\begin{aligned}
\text{Var}(y_p - \hat{y}_p) &= \text{Var}(a_p^\top \beta + \varepsilon_p - a_p^\top \hat{\beta}) = \text{Var}(\varepsilon_p - a_p^\top \hat{\beta}) \\
&= \text{Var}(\varepsilon_p) + \text{Var}(a_p^\top \hat{\beta}) \\
&= \sigma^2 + \sigma^2 a_p^\top (X^\top X)^{-1} a_p \\
&= \sigma^2 (1 + a_p^\top (X^\top X)^{-1} a_p).
\end{aligned}$$

So, for known σ^2 we have that,

$$\frac{y_p - \hat{y}_p}{\sigma \sqrt{1 + a_p^\top (X^\top X)^{-1} a_p}} \sim N(0,1).$$

However, since we rarely known σ^2 we substitute with the estimator for sample variance $S^2 = \frac{SSE}{N-p-1}$ instead of σ^2 so we get that,

$$\frac{y_p - \hat{y}_p}{S \sqrt{1 + a_p^\top (X^\top X)^{-1} a_p}} \sim t_{N-p-1}.$$

□

10 Lecture 10: Demo (redacted)

11 Lecture 11: Analysis of Variance

From our Multiple Linear Regression framework we have $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i$. Recall that our Sum of Squared Errors can be written as $\text{SSE}(\hat{\beta}) = \sum_{i=1}^{N} r_i^2 = r^\top r = (Y - X\hat{\beta})^\top (Y - X\hat{\beta})$.

11.1 Hypothesis Test

Test the null hypothesis $H_0 : \beta_1 = \beta_2 = \cdots = 0$ and reject H_0 if $H_{0i} : \beta_i = 0$ for $i = 1, \ldots, p$ for $\alpha = 0.05$. Assuming that H_0 is true and $\hat{\beta}_i$'s are independent,

$P(\text{reject } H_0 \mid H_0 \text{ is true.}) = P(\text{reject any } H_{0i} \text{ for } i = 1, \ldots, p \mid H_0) = 1 - P(\text{accept all } H_{0i} \mid H_0) = 1 - 0.95^p$.

11.2 Reduced/Restriced Model

Under H_0 our response $y_i = \beta_0 + \varepsilon_i$,

(i) **LSE**: $\hat{\beta}_0 = \bar{y}$

(ii) $\text{SSE}(\hat{\beta}_0) = \sum_{i=1}^{N} (y_i - \bar{y})^2 = \text{SST}$

(iii) $\text{SSE}(\hat{\beta}_0) - \text{SSE}(\hat{\beta}) = \text{SST} - \text{SSE} = \text{SSR}$

$$
\begin{aligned}
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 = (y - X\hat{\beta})^\top (y - X\hat{\beta}) = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2 - [(I-H)y]^\top [(I-H)y] \\
&= y^\top y - n\bar{y}^2 - y^\top (I-H)^\top (I-H) y \qquad I - H \text{ idempotent + symmetric} \\
&= y^\top y - n\bar{y}^2 - y^\top (I-H) y \\
&= y^\top H^\top H y - n\bar{y}^2 \qquad H \text{ idempotent and symmetric} \\
&= y^\top H H y - n\bar{y}^2 \\
&= \hat{\beta}^\top X^\top X \hat{\beta} - n\bar{y}^2
\end{aligned}
$$

Under $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ we have that $\dfrac{SSR}{\sigma^2} \sim \chi_p^2$ with F-statistic defined as,

$$F = \frac{\text{SSR}/p}{\text{SSE}/N - p - 1} = \frac{MSR}{MSE} \sim F_{p, N-p-1}.$$

Source of Variation	**df**	**Sum of Squares**	**Mean Squares**	**F**
Regression	p	$\text{SSR} = \hat{\beta}^\top X^\top X \hat{\beta} - n\bar{y}^2$	$\text{MSR} = \dfrac{\text{SSR}}{p}$	$\dfrac{\text{MSR}}{\text{MSE}}$
Residual	$n - p - 1$	$\text{SSE} = (y - X\hat{\beta})^\top (y - X\hat{\beta})$	$\text{MSE} = \dfrac{\text{SSE}}{n - p - 1}$	
Total	$n - 1$	$\text{SST} = \sum_{i=1}^{n} (y_i - \bar{y})^2$		

Table 2: ANOVA Table for Linear Regression.

Remark 11.1. The total coefficient of determination is given by $R^2 = \dfrac{SSR}{SST}$ for $0 \leq R^2 \leq 1$.

11.3 Likelihood Estimation

Since $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i \sim N(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}, \sigma^2)$ independently, we have that the pdf of y is,

$$f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{(y-\mu)^2}{2\sigma^2} \right\}.$$

Thus the likelihood function can we given as,

$$f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{(y-\mu)^2}{2\sigma^2} \right\}$$

So our log-likelihood function is derived as,

$$\begin{aligned}
L(\beta, \sigma^2) &= \prod_{i=1}^{n} f(y_i) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{\left[y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})\right]^2}{2\sigma^2} \right\} \\
\ell(\beta, \sigma^2) &= \log L(\beta, \sigma^2) \\
&= \sum_{i=1}^{n} \left[\log(2\pi\sigma^2)^{-\frac{1}{2}} - \frac{\left[y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})\right]^2}{2\sigma^2} \right] \\
&= -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left[y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})\right]^2.
\end{aligned}$$

Maximizing the log-likelihood function $l(\beta, \sigma^2)$ is equivalent to minimizing the residual sum of squares. Thus,

$$\hat{\beta}_{\text{MLE}} = \hat{\beta}_{\text{LSE}} \quad \text{and} \quad \text{RSS}(\hat{\beta}) = \text{SSE}.$$

If we want to set $\frac{\partial l(\hat{\beta}, \sigma^2)}{\partial \sigma^2} = 0$ so that,

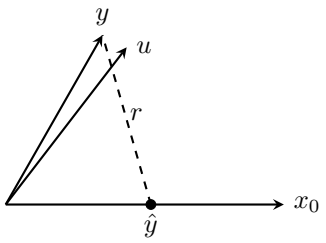
$$\frac{\partial \ell(\hat{\beta}, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2} \cdot \frac{1}{2\pi\sigma^2} \cdot 2\pi \; - \; \frac{\text{SSE}}{2} \Big(- \frac{1}{(\sigma^2)^2} \Big) = 0$$

which means that $\dfrac{SSE}{\sigma^2} = n$ so $\hat{\sigma}^2 = \frac{SSE}{n}$. Note that the MLE does not correct for bias from estimating $\beta_0, \ldots, \beta_p$ therefore losing degrees of freedom making it an underestimate i.e., biased.

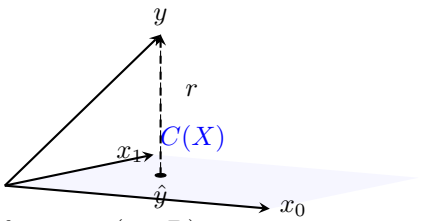
12 Lecture 12: Geometric Interpretation of Least Squares

Geometricaally, we can view X as a column space $C(X)$ which makes it a set of all vectors that can be constructed as a linear combination of the columns of X. So $\hat{Y} = X\hat{\beta} \in C(X)$. Since $C(X)$ spans a $p+1$ dimensional subspace in the n-dimensional space we can see that $\beta_0 X_0 + \beta_1 X_1$ can span a plane.

Since least squares minimizes the squared residuals i.e. $\sum_{i=1}^{N} r_i^2$ for residual vector r, then we can also say that the length of r i.e. the norm $\sqrt{r_1^2 + \cdots + r_N^2}$ is subject to the constraint that $\hat{Y} \in C(X)$ since $Y = \hat{Y} + r$. In other words, least squares looks for $\hat{Y} \in C(X)$ that is closest to Y,



for $p = 0$



for $p = 1$ (in 3D)

Observe that $r \perp C(X)$ which also implies that $1^\top r = 0, X_1^\top r = 0, \ldots, X_p^\top r = 0$. We can represent these p orthogonal features from our feature matrix to the residual matrix as,

$$X^\top r = 0 \in \mathbb{R}^{(p+1)\times 1} \implies X^\top (Y - X\hat{\beta}) = 0 \implies X^\top X\hat{\beta} = X^\top Y \implies \hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Theorem 12.1.

(i) Y is projected into 2 orthogonal subapces $C(X)$ and $C^\perp(X)$ where $\hat{Y} \in C(X)$ and $r \in C^\top(X)$, with $p+1$ and $N-p-1$ dimensions respectively.

(ii) Under normality, we have that $\hat{Y}$ and r are independent.

12.1 Testing Linear Constraints of β_i's

(i) For $H_0 : \beta_q = \beta_{q+1} = \cdots = \beta_p = 0$ with $1 \le q \le p$ we can re-write H_0 as $A\beta = 0$,

$$A = \underbrace{\begin{bmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 1 \end{bmatrix}}_{(p-q+1)\times(p+1)}.$$

(ii) Model $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$ and we want to enforce $\beta_1 = 2\beta_2$ and $\beta_3 = 0$.

$$A = \begin{bmatrix} 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \iff A\beta = 0.$$

(iii) Suppose that $\beta_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$. We then have a linear constraint by $\beta_1 = \beta_2$. Then $\hat{Y}$ is the orthogonal projection of Y onto $C(X)$ and $\hat{Y}_A$ is the orthogonal projection of Y onto $C_A(X)$. If $H_0 : A\beta = 0$ is true, so we expect that $\hat{Y}$ and $\hat{Y}_A$ to be close i.e., the squared distance is $\|\hat{Y} - \hat{Y}_A\|^2 = (\hat{Y} - \hat{Y}_A)^\top (\hat{Y} - \hat{Y}_A)$.

Remark 12.1. We use $A\beta = 0$ to enforce constraints.

13 Lecture 13: Sum of Squares and Additional Distance

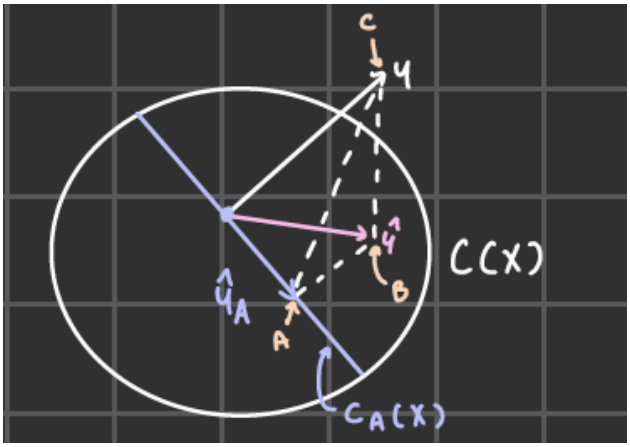


Figure 3: Squared Distance.

13.1 Testing Framework: $A\beta = 0$

We test $H_0 : A\beta = 0$ for $A \in \mathbb{R}^{l\times(p+1)}$.
There are a couple of useful identities for computing the squared distance notably, $(AC)^2 = \text{SSE}_A$ and $(BC)^2 = \text{SSE}$. Thus, we have that,

$$\|\hat{Y} - \hat{Y}_A\|^2 = (AB)^2 = (AC)^2 - (BC)^2 = \text{SSE}_A - \text{SSE}.$$

Theorem 13.1. *Under $H_0 : A\beta = 0$ where $A \in \mathbb{R}^{l\times(p+1)}$ matrix,*

(i) $\dfrac{\|\hat{Y} - \hat{Y}_A\|^2}{\sigma^2} \sim \chi^2_l$.

(ii) $\|\hat{Y} - \hat{Y}_A\|^2$ *is independent of* $\sigma^2 = \dfrac{(Y - \hat{Y})^\top (Y - \hat{Y})}{N - p - 1}$.

(i) and (ii) imply that $F = \dfrac{\|\hat{Y} - \hat{Y}_A\|/l}{\sigma^2} \sim F(l, N - p - 1)$.

Thus, we reject H_0 at level α if $F > F_\alpha(l, N - p - 1)$ where the p-value is $P(F(l, N - p - 1) > F)$.

Remark 13.1. The general linear hypothesis $H_0 : A\beta = 0$ compares the full model to the restricted model obtained by imposing the constraint. If the restriction substantially worsens the fit (i.e., increases SSE), we reject H_0; if the restriction does not worsen the fit appreciably, we fail to reject.

Remark 13.2. To test $H_0 : A\beta = 0$ we,

(i) Fit the full model without restriction

(ii) Compute SSE.

(iii) Fit restricted model with $A\beta = 0$

(iv) Compute SSE_A

(v) Compute $F = \dfrac{(\text{SSE}_A - \text{SSE})/l}{\text{SSE}/N - p - 1}$.

Example 13.1. Suppose we wish to model $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$ and we want to enforce the constraint the $\beta_1 = 2\beta_2$ and $\beta_3 = 0$. Then our matrix becomes,

$$A = \begin{bmatrix} 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

So $A\beta = 0$. Then we can write our full model and restricted models as,

$$\begin{aligned} y &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon \\ y_A &= \beta_0 + \beta_2(2X_1 + X_2) + \varepsilon, \end{aligned}$$

where $X_4 = 2X_1 + X_3$.

14 Lecture 14: Interaction, Model Specification, and Piecewise Regression

14.1 UFFI Example and Interpretation

Definition 14.1. We consider a regression model for CH_2O concentration with predictors: UFFI presence (indicator), air tightness x, and their interaction. The fitted means $\mu_i = X_i^\top \beta$ for different settings of UFFI and air tightness are:

UFFI	Air tightness	$X^\top$	$\mu_i = X_i^\top \beta$
Present	$x+1$	$(1,1,x+1)$	$\beta_0 + \beta_1 + \beta_2(x+1)$
Present	x	$(1,1,x)$	$\beta_0 + \beta_1 + \beta_2 x$
Absent	$x+1$	$(1,0,x+1)$	$\beta_0 + \beta_2(x+1)$
Absent	x	$(1,0,x)$	$\beta_0 + \beta_2 x$

These four rows allow clean comparisons of mean responses when UFFI or air tightness changes. Below we interpret the differences algebraically.

(i) Row 4 − Row 2:

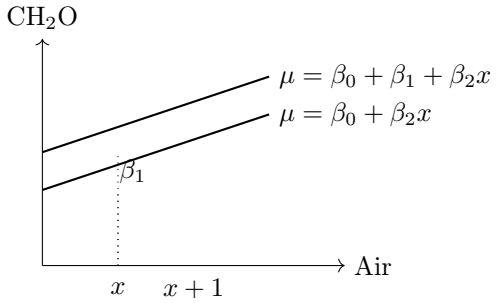
$$\beta_0 + \beta_2 x = \beta_0 \quad \text{if } x = 0$$

Thus β_0 represents the expected CH_2O level when UFFI is absent and air tightness is zero.

(ii) Row 2 − Row 4 = β_1 Interpreted as the expected increase in CH_2O when UFFI becomes present (holding air level at x).

(iii) Row 3 − Row 4 = β_2 This is the expected increase in CH_2O when air tightness increases by one unit with UFFI absent.

(iv) Row 1 − Row 2 = β_2 Same interpretation, but now for the UFFI-present group. This matches the assumption of *additivity* when no interaction term is present.



This picture assumes the UFFI and air effects are additive; this may or may not be true in practice.

14.2 Interaction and Model Specification

When additivity is questionable, we use the full model with interaction:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i,$$

where $x_{3i} = x_{1i}x_{2i}$ encodes the interaction. Interactions allow the slope of x_2 to differ depending on x_1.

Definition 14.2. Interaction. The influence of one predictor on the response changes depending on the level of another predictor.

Testing for a UFFI effect. If UFFI has no effect, neither its main effect nor its interaction should matter:

$$H_1 : \beta_1 = \beta_3 = 0.$$

Under this restriction, the reduced model is:

$$y_i = \beta_0 + \beta_2 x_{2i} + \varepsilon_i,$$

meaning air tightness alone determines CH_2O concentration.

14.3 Quadratic and Piecewise Models

Quadratic regression allows curvature:

$$E(y) = \beta_0 + \beta_1 x + \beta_2 x^2.$$

Matrix representation:

$$E(Y) = X\beta, \quad X = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{pmatrix}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

Piecewise constant model.

$$y_i = \beta_0 I(x < a) + \beta_1 I(x \geq a),$$

which represents a discontinuous jump at a. Design matrix:

$$X = \begin{pmatrix} 1 & I(x_1 \geq a) \\ 1 & I(x_2 \geq a) \\ \vdots & \vdots \\ 1 & I(x_n \geq a) \end{pmatrix}.$$

Piecewise linear model. This allows both slope and intercept to change at a. Naive writing:

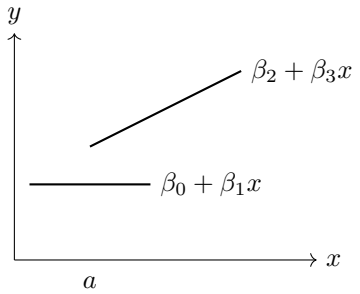
$$y = \begin{cases} \beta_0 + \beta_1 x, & x < a, \\ \beta_0 + \beta_2 x, & x \geq a. \end{cases}$$

To express this within a linear model:

$$y_i = \beta_0 I(x_i < a) + \beta_1 x_i I(x_i < a) + \beta_2 I(x_i \geq a) + \beta_3 x_i I(x_i \geq a).$$

Design matrix:

$$X = \begin{pmatrix} 1 & x_1 I(x_1 < a) & I(x_1 \geq a) & x_1 I(x_1 \geq a) \\ 1 & x_2 I(x_2 < a) & I(x_2 \geq a) & x_2 I(x_2 \geq a) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n I(x_n < a) & I(x_n \geq a) & x_n I(x_n \geq a) \end{pmatrix}.$$



15 Lecture 15: Piecewise Linear and Continuous

15.1 Piecewise Linear and Continuous

Model:

$$E(y) = \begin{cases} \beta_0 + \beta_1 x, & x < a, \\ \beta_2 + \beta_3 x, & x \geq a. \end{cases}$$

with constraint

$$\beta_0 + \beta_1 a = \beta_2 + \beta_3 a \quad \Longleftrightarrow \quad \beta_2 = \beta_0 + (\beta_1 - \beta_3)a.$$

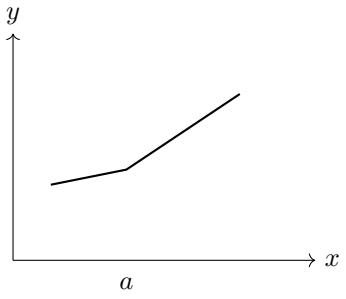
So new model becomes:

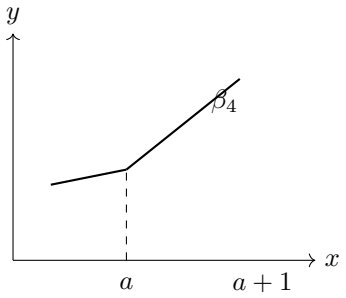
$$E(y) = \begin{cases} \beta_0 + \beta_1 x, & x < a, \\ \beta_0 - (\beta_1 - \beta_3)a + \beta_3 x, & x \geq a, \end{cases}$$

i.e.

$$E(y) = \beta_0 + \beta_1 a + \beta_3 (x - a).$$

$$\begin{aligned} E(y) &= \beta_0 I(x < a) + \beta_1 x I(x < a) + (\beta_0 + \beta_1 a) I(x \geq a) + \beta_3 (x - a) I(x \geq a) \\ &= \beta_0 + \beta_1 x - \beta_1 x I(x \geq a) + \beta_1 a I(x \geq a) + \beta_3 (x - a) I(x \geq a) \\ &= \beta_0 + \beta_1 x - \beta_1 (x - a) I(x \geq a) + \beta_3 (x - a) I(x \geq a) \\ &= \beta_0 + \beta_1 x + (\beta_3 - \beta_1)(x - a) I(x \geq a) \\ &= \beta_0 + \beta_1 x + \beta_4 (x - a) I(x \geq a). \end{aligned}$$





15.2 One Sample Problem

$$y_i = \beta_0 + \varepsilon_i, \qquad E(y_i) = \beta_0.$$

$$Y = X\beta, \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}, \quad \beta = \beta_0.$$

15.3 Two Sample Problem: "Cell Means" Model

$$y_{ij} = \mu_i + \varepsilon_{ij}, \qquad i = 1, 2, \qquad \text{each group has } n_i \text{ observations.}$$

$$E(Y) = X\beta$$

$$Y = \begin{pmatrix} y_{11} \\ \vdots \\ y_{1n_1} \\ y_{21} \\ \vdots \\ y_{2n_2} \end{pmatrix}, \qquad E(Y) = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_1 \\ \mu_2 \\ \vdots \\ \mu_2 \end{pmatrix} = \begin{pmatrix} 1_{n_1\times 1} & 0_{n_1\times 1} \\ 0_{n_2\times 1} & 1_{n_2\times 1} \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}.$$

Thus,

$$X = \begin{pmatrix} 1_{n_1\times 1} & 0_{n_1\times 1} \\ 0_{n_2\times 1} & 1_{n_2\times 1} \end{pmatrix}, \qquad \beta = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}.$$

Compute:

$$X^TX = \begin{pmatrix} n_1 & 0 \\ 0 & n_2 \end{pmatrix}.$$

$$(X^TX)^{-1}X^TY = \begin{pmatrix} \frac{1}{n_1}\sum_{j=1}^{n_1} y_{1j} \\ \frac{1}{n_2}\sum_{j=1}^{n_2} y_{2j} \end{pmatrix} = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \end{pmatrix}.$$

15.4 Effects Model

$$E(y_i) = \beta_0 + \beta_1 x_i, \qquad x_i = I(\text{observation is in group 2}).$$

$$E(Y) = \begin{pmatrix} y_1 \\ \vdots \\ y_{n_1} \\ y_{n_1+1} \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1_{n_1\times 1} & 0_{n_1\times 1} \\ 1_{n_2\times 1} & 1_{n_2\times 1} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

$$X^TX = \begin{pmatrix} n_1 + n_2 & n_2 \\ n_2 & n_2 \end{pmatrix}.$$

Given

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

we compute

$$(X^TX)^{-1} = \frac{1}{n_1 n_2}\begin{pmatrix} 1 & -1 \\ -1 & \frac{n_1+n_2}{n_2} \end{pmatrix}.$$

$$X^TY = \begin{pmatrix} \sum_{i=1}^{n_1+n_2} y_i \\ \sum_{j=1}^{n_2} y_{2j} \end{pmatrix}.$$

Then

$$\hat{\beta} = (X^TX)^{-1}X^TY = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 - \bar{y}_1 \end{pmatrix}.$$

This can be used for UFFI regression model 1.

16 Lecture 16: k-Sample Problem, ANOVA and Multicollinearity

16.1 k-Sample Problem

Group	Observations
1	$y_{11}, y_{12}, \ldots, y_{1n_1}$
2	$y_{21}, y_{22}, \ldots, y_{2n_2}$
$\vdots$	$\vdots$
k	$y_{k1}, y_{k2}, \ldots, y_{kn_k}$

16.2 "Cell Means" Model

$$y_{ij} = \mu_i + \varepsilon_{ij}, \qquad i = 1, \ldots, k.$$

$$E(Y) = \begin{pmatrix} y_{11} \\ \vdots \\ y_{1n_1} \\ y_{21} \\ \vdots \\ y_{2n_2} \\ \vdots \\ y_{k1} \\ \vdots \\ y_{kn_k} \end{pmatrix} = \underbrace{\begin{pmatrix} \mathbf{1}_{n_1 \times 1} & 0 & \cdots & 0 \\ 0 & \mathbf{1}_{n_2 \times 1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{1}_{n_k \times 1} \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_k \end{pmatrix}}_{\beta}.$$

16.3 Matrix Calculations

$$X^T X = \begin{pmatrix} n_1 & 0 & \cdots & 0 \\ 0 & n_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & n_k \end{pmatrix} = \operatorname{diag}(n_1, \ldots, n_k).$$

$$X^T Y = \begin{pmatrix} \sum_{j=1}^{n_1} y_{1j} \\ \sum_{j=1}^{n_2} y_{2j} \\ \vdots \\ \sum_{j=1}^{n_k} y_{kj} \end{pmatrix}.$$

Thus

$$\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{pmatrix} \frac{1}{n_1} \sum_{j=1}^{n_1} y_{1j} \\ \frac{1}{n_2} \sum_{j=1}^{n_2} y_{2j} \\ \vdots \\ \frac{1}{n_k} \sum_{j=1}^{n_k} y_{kj} \end{pmatrix} = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_k \end{pmatrix}.$$

Fitted values:

$$\hat{y} = X\hat{\beta} = \begin{pmatrix} \hat{y}_{11} \\ \vdots \\ \hat{y}_{1n_1} \\ \vdots \\ \hat{y}_{kn_k} \end{pmatrix} = \begin{pmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_2 \\ \vdots \\ \bar{y}_k \\ \vdots \\ \bar{y}_k \end{pmatrix}.$$

$$y_{i+} = \sum_{j=1}^{n_i} y_{ij}.$$

16.4 Effects Model

$$E(y_i) = \beta_1 + \beta_2 x_{i2} + \cdots + \beta_k x_{ik}, \qquad x_{ij} = I(\text{obs. is in group } j).$$

Equivalently,

$$E(Y) = \begin{pmatrix} 1 & x_{12} & \cdots & x_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n2} & \cdots & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix}.$$

Reference group is the one with all indicator variables equal to 0. It can be shown:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

and group mean differences satisfy

$$\hat{\beta}_j = \bar{y}_j - \bar{y}_1, \qquad j = 2, \ldots, k.$$

16.5 ANOVA

Sum of squares for regression:

$$\begin{aligned} SSR &= \hat{\beta}^T X^T \hat{Y} - n\bar{y}^2 \\ &= \hat{\beta}^T X^T Y - n\bar{y}^2 \\ &= \sum_{i=1}^{k} n_i (\bar{y}_i - \bar{y})^2. \end{aligned}$$

Sum of squared errors:

$$SSE = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2.$$

16.6 ANOVA Table

Source	Sum of Squares	*df*	Mean Squares
Regression	$\sum_{i=1}^{k} n_i(\bar{y}_i - \bar{y})^2$	$k-1$	$MSR = \frac{SSR}{k-1}$
Residual	$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_i)^2$	$n-k$	$MSE = \frac{SSE}{n-k}$
Total	$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y})^2$	$n-1$	

F-statistic:

$$F = \frac{MSR}{MSE}.$$

16.7 Multicollinearity

Perfect Multicollinearity exists when there are linearly dependent columns of a design matrix $1, x_1, x_2, \ldots, x_p$ i.e., there exists constants c_i for $i = 0, \ldots, p$ such that $c_0 1 + c_1 X_1 + \cdots + c_p X_p = 0$.
Multicollinearity in general is defined as the existence of constants c_i for $i = 0, \ldots, p$ such that $c_0 1 + c_1 X_1 + \cdots c_p X_p = 0$ but may not be exactly linearly dependent.

16.8 Consequences of Multicollinearity

Perfect Multicollinearity amongst columns of our design matrix X makes the determinant of the hat matrix $|X^\top X| = 0$ so the inverse of the hat matrix does not exist. Thus, our least squares estimates $\hat{\beta} = (X^\top X)^{-1} X^\top Y$ does not exist either.

Theorem 16.1. ***Effects of Perfect Multicollinearity.*** *The columns of the feature matrix X are said to be perfectly multicollinear if there exist constants $c_0, c_1, \ldots, c_p$, not all zero, such that*

$$c_0 \mathbf{1} + c_1 X_1 + \cdots + c_p X_p = 0.$$

In this case the matrix $X^\top X$ is singular, i.e. $\det(X^\top X) = 0$*, so $(X^\top X)^{-1}$ does not exist and the least squares estimator $\hat{\beta}$ is not uniquely defined. Moreover, when the predictors are nearly multicollinear (i.e.* high *multicollinearity), the matrix $X^\top X$ becomes close to singular, so the entries of $(X^\top X)^{-1}$ become very large. Since*

$$\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1},$$

this implies that $\mathrm{Var}(\hat{\beta}_i)$ *becomes very large, making coefficient estimates unstable.*

16.9 Issues with Multicollinearity

(i) Variance of least-squares estimates $\hat{\beta}$ are large

(ii) Important predictors become insignificant in the model

(iii) It is hard to distinguish the effects $\implies$ interpretation of coefficients is problematic

16.10 Detecting Multicollinearity

16.10.1 First Check

Pairwise Sample Correlation Coefficient

$$r_{\ell m} = \frac{\sum_{i=1}^{n}(x_{i\ell} - \bar{x}_\ell)(x_{im} - \bar{x}_m)}{\sqrt{\sum_{i=1}^{n}(x_{i\ell} - \bar{x}_\ell)^2 \quad \sum_{i=1}^{n}(x_{im} - \bar{x}_m)^2}}$$

If $|r_{lm}| \approx 1$ then predictors X_l and X_m are strongly linearly related,
so there is not need to have *both* in the model.

16.10.2 Variance Inflation Factor (VIF)

(i) X_k is regressed on all other $p-1$ X's so,

$$X_{ik} = \beta_0^* + \beta_1^* X_{i1} + \cdots \beta_{k-1}^* X_{i,k-1} + \beta_{k+1}^* x_{i,k+1} + \cdots + \beta_p^* x_{ip} + \varepsilon_i.$$

(ii) $R_i^2 = \dfrac{SSR}{SST}$ where R_k^2 measures how strongly X_k is linearly related to the rest of the X's

(iii) Then, $VIF_k = \dfrac{1}{1 - R_k^2}$ where,

(a) $VIF_k > 10$ suggests strong evidence of multicollinearity

(b) $VIF_k > 5$ suggests some evidence of multicollinearity

17 Lecture 17: Model Checking

17.1 Examination of Residuals

To check a model, we can look at the residuals $r_i = y_i - \hat{y}_i$ for corresponding response y_i i.e., the difference between observed and fitted y. So, under regular assumptions and if $n >> p$ (number of data points is larger than the number of coefficients β), then r_i is approximately ε_i.

Theorem 17.1. *If $n >> p$ then $r_i \approx \epsilon_i$ for small $H = X(X^\top X)^{-1}X^\top$.*

Proof. Let $r = (I - H)\varepsilon$, then

$$\begin{aligned} r &= Y - \hat{Y} \\ &= Y - HY = (I - H)Y \\ &= (I - H)Y = (I - H)(X\beta + \varepsilon) \\ \implies (I - H)X\beta &= X\beta - X(X^\top X)^{-1}X^\top X\beta = X\beta - X\beta = 0. \end{aligned}$$

i.e., for small H we have that $r \approxeq \varepsilon$. □

Theorem 17.2. ***Properties of*** *H*

(i) $H^2 = H$ implies that the ith diagonals $h_{ii} = (H)_{ii} = \sum_{i=1}^{N} h_{ij}h_{ji}$.

Proof.

$$\begin{aligned} &= \sum_{j=1}^{n} h_{ij}^2 = h_{ii}^2 + \sum_{j=1}^{N} h_{ij}^2 \implies \sum_{j\neq i} h_{ij}^2 = h_{ii}(1 - h_{ii}) \\ \implies 0 \le h_{ii} \le 1 &\implies 0 \le h_{ii}(1 - h_{ii}) \le \frac{1}{4}. \end{aligned}$$

We have established a *bound* on how large the diagonals of H can be.
Thus the off-diagonal elements of H cannot be large. □

(ii) $\sum_{i=1}^{N} h_{ii} = tr(H) = tr(X(X^\top X)^{-1}X^\top) = tr((X^\top X)^{-1}X^\top X) = tr(I_{p+1}) = p + 1.$
Thus from the above, we see that the average of the h_{ii} values is $\frac{p+1}{n}$.
If $n >> p$ then H will be small, so $r = (I - H)\varepsilon \approx \varepsilon$.
If $n = p + 1$ then the average of h_{ii} is 1 thus $h_{ii} = 1, h_{ij} = 0$ for all $i \neq j$.
Thus, $H = I \implies \hat{Y} = HY = Y \implies r = 0$.

17.2 Standardized & Studentized Residuals

Recall that for residuals $r = (I-H)\varepsilon$ we had that $\mathbb{E}[r] = 0$ and $\mathrm{Var}(r) = \sigma^2(I-H)$ with $r \sim \mathrm{MVN}(0, \sigma^2(I-H))$. What if $Y \neq X\beta + \varepsilon$? Say instead it's $Y = X\beta + Z\alpha + \varepsilon$ but we used the fitted model $Y = X\beta + \varepsilon$. On the last page, we showed that $(I - H)X\beta = 0$, so similarly for our new model,

$$\mathbb{E}[r] = (I - H)\mathbb{E}[Y] = (I - H)(X\beta + Z\alpha) = (I - H)Z\alpha = 0.$$

(i) **Standardized Residuals**: $r_i^2 = \dfrac{r_i}{\hat{\sigma}}$ for $i = 1, \ldots, N$.

(ii) **Studentized Residuals**: $d_i = \dfrac{r_i}{\sqrt{\hat{\sigma}^2(1 - h_{ii})}}$ for $i = 1, \ldots, N$.

Remark 17.1. Under assumptions of random errors, $d_1, \ldots, d_N$ are approximately $N(0, 1)$.

17.2.1 Residual Checking Plots

Check $\mathbb{E}[\varepsilon] = 0$: We can look for likely causes for violations of $\mathbb{E}[\varepsilon_i] = 0$,

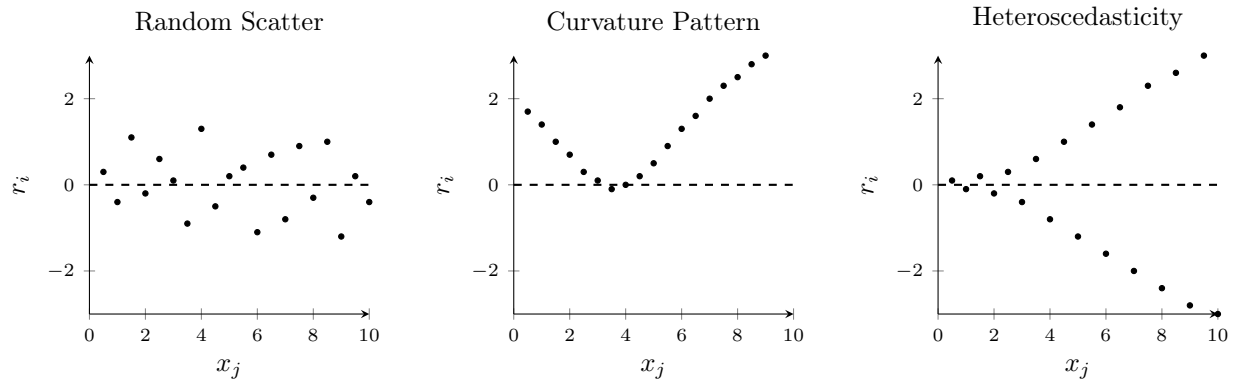
(i) Effect of predictors of the response variables is not in fact linear

(ii) Omission of some important predictors

3 Plots of checks: (1) residuals vs. x_j for $j = 1, \ldots, p$, (2) residuals vs. $\hat{y}$, (3) added variable plots.

17.2.2 Residuals vs X_j

$r_i = y_i - (\hat{\beta}_0 - \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip})$.
Plot $r_1, \ldots, r_N$ against $x_{1j}, \ldots, X_{nj}$. Then we expect random scatter for $j = 1, \ldots, p$.
No special trends *if* points fall into a horizontal band around 0.
Non-random pattern suggests non-linearity so we can adapt the way X_j is modelled which may require higher order terms like X_j^2 suggesting the use of polynomial regression.



17.2.3 Residuals vs. $\hat{y}$

If the model is adequate i.e., $\mathbb{E}[\varepsilon] = 0$ then $\mathrm{Cov}(r, \hat{Y}) = 0$.

Proof.

$$\mathrm{Cov}(r, \hat{y}) = \mathrm{Cov}\big((I - H)Y,\ HY\big) = (I - H)\,\mathrm{Cov}(Y)\,H^\top = (I - H)\,\sigma^2 I\,H^\top = \sigma^2 (I - H)H^\top.$$

□

The residuals should lie within a horizontal band around zero with no special pattern.

17.2.4 Studentized Residuals vs. $\hat{y}$

Instead of r_i vs. $\hat{y}$ we can plot d_i vs $\hat{y}$. If the model is adequate then,

(i) d_i lies within a horizontal band around zero and should not exhibit any special pattern

(ii) Approximately 95% of the d_i's should be within a $(-2, 2)$ since $N(0, 1)95\%$ CI is ± 1.96, and almost all of them should be within $(-3, 3)$.

18 Lecture 18: Checking Plots cont., and Influence

18.1 Added-Variable Plots

We suspect that an important predictor X^* has not been included in our model but is part of the data.

(i) Regress y on all current predictors $X_1, \ldots, X_p$ to get the residual vector r

(ii) Regress X^* on all $X_1, \ldots, X_p$ to get residual vector t

(iii) Plot r vs. t. If we observe a pattern, then X^* may be a useful addition to the model.

18.2 Check $\mathrm{Var}(\varepsilon) = \sigma^2$

Use r_i vs. $\hat{y}_i$ to detect non-constant variance (heteroscedasticity) and check if the residuals seem to have constant variability wrt fitted values (no patterns).

18.3 Check Correlation

This can prove to be difficult but can be done if there is some structure to the data. For example, temporal/spatial relationship violations of independence (having correlation) can be observed and lead to incorrect estimation of the variance.

18.3.1 Correlation across time

Example 18.1. For any two observations t time units apart, we have that the correlation $\rho_t = \mathrm{Corr}(\varepsilon_i, \varepsilon_{i-1}) = \rho^t$ for $t \geq 0$. For example,

1. For $\rho \approx 0.9$, then we will observe a smooth residual trend with long runs above or below zero.
2. For $\rho \approx 0$, then we will observe the residuals jumping randomly with no visible pattern.
3. For $\rho \approx -0.9$, then the residuals will alternate up and down frequently.

18.4 Durbin-Watson Statistic

Definition 18.1. Durbin–Watson Statistic. Given residuals $r_1, \ldots, r_N$ from a fitted linear regression model, the Durbin–Watson statistic is defined as

$$d \;=\; \frac{\displaystyle\sum_{i=2}^{N} (r_i - r_{i-1})^2}{\displaystyle\sum_{i=1}^{N} r_i^2}.$$

Under the usual first-order autoregressive model for the residuals,

$$r_i = \rho\, r_{i-1} + \varepsilon_i, \qquad |\rho| < 1,$$

the Durbin–Watson statistic satisfies the approximation

$$d \;\approx\; 2(1 - \rho).$$

Interpretation.

- $d \approx 2$ indicates *no first-order autocorrelation* ($\rho \approx 0$).
- $d < 2$ indicates *positive* autocorrelation ($\rho > 0$).
- $d > 2$ indicates *negative* autocorrelation ($\rho < 0$).

Hypothesis Tests. The Durbin–Watson test evaluates the null hypothesis of no first-order autocorrelation:

$$H_0 : \rho = 0 \qquad \text{vs.} \qquad H_1 : \rho \neq 0.$$

For one-sided alternatives, we consider:

$$\begin{aligned} &\text{Positive autocorrelation:} \quad H_1 : \rho > 0 \iff d < 2, \\ &\text{Negative autocorrelation:} \quad H_1 : \rho < 0 \iff d > 2. \end{aligned}$$

Critical regions depend on N and p via Durbin–Watson lower and upper bounds (d_L, d_U). For example, for testing *positive* autocorrelation:

$$\begin{cases} d < d_L & \Rightarrow \text{Reject } H_0 \text{ (evidence of positive autocorrelation)}, \\ d > d_U & \Rightarrow \text{Fail to reject } H_0, \\ d_L \leq d \leq d_U & \Rightarrow \text{Test is inconclusive.} \end{cases}$$

What the Statistic Tells Us. The Durbin–Watson statistic measures the degree of *systematic dependence between successive residuals*. Detecting autocorrelation is crucial because:

- it violates the standard linear model assumption $\text{Cov}(r_i, r_j) = 0$ for $i \neq j$;
- OLS estimates remain unbiased, but the estimated variances of $\hat{\beta}$ are incorrect;
- t-tests and F-tests become unreliable if autocorrelation is present.

Thus, the Durbin–Watson statistic acts as a diagnostic for whether the regression model correctly captures time-ordered structure (or omitted variables) that create correlated errors.

18.5 Outliers

We define a particular case with an unusual value in y or x as an outlier. These often have significant effects on the fitted regression euqation so it is important to determine whether they should be retained or eliminated.

18.5.1 Detect Outliers

Given the studentized residuals $d_i = \dfrac{r_i}{\sqrt{\hat{\sigma}^2(1 - h_{ii})}}$ then large d_i values indicate outliers in y. Recall that from our hat matrix $H = X(X^\top X)^{-1}X^\top$ where $\hat{Y} = HY$ and our response $\hat{y}_i = \displaystyle\sum_{i=1}^{N} h_{ij}y_j = h_{ii}y_i = \sum_{j \neq i}^{N} h_{ij}y_j$,

- h_{ii} represents how much y_i contributes to $\hat{y}_i$
- large h_{ii} vs. other h_{ij}'s imply that h_{ij} dominates $\hat{y}_i$
- by (ii) the fitted line must pass close to the ith observation since i has high leverage on the fitted line
- As $h_{ii} \to 1$ then $\hat{y}_i \to y_i$.

18.5.2 Leverage of the ith Observation: h_{ii}

(i) The leverage h_{ii} reflects how far the ith predictor vector

$$(x_{i1}, \ldots, x_{ip})$$

lies from the center of the cloud of all predictor values.

(ii) Leverage is *small* when $(x_{i1}, \ldots, x_{ip})$ is close to the centroid

$$(\bar{x}_1, \ldots, \bar{x}_p),$$

and *large* when the observation is far from this centroid.

(iii) In simple linear regression ($p = 1$), the leverage reduces to

$$h_{ii} = \frac{1}{N} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

- Leverage is minimized when $x_i = \bar{x}$, and increases as x_i moves away from $\bar{x}$.
- Since $0 \le h_{ii} \le 1$ for all i, and

$$\sum_{i=1}^{N} h_{ii} = p + 1,$$

a common diagnostic rule is:

$$h_{ii} > \frac{2(p+1)}{N}$$

which flags potential high-leverage (influential) observations in the predictor space.

Proof. For $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$,

$$\begin{aligned}
X &= \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \\
X^\top X &= \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}, \\
(X^\top X)^{-1} &= \frac{1}{n\sum x_i^2 - (\sum x_i)^2} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}, \\
h_i &= [1 \;\; x_i]\,(X^\top X)^{-1} \begin{bmatrix} 1 \\ x_i \end{bmatrix} \\
&= \frac{1}{n\sum x_i^2 - (\sum x_i)^2} \left(\sum x_i^2 - 2x_i \sum x_i + n x_i^2 \right) \\
&= \frac{1}{nS_{xx}} \left(n x_i^2 - 2x_i n\bar{x} + n\bar{x}^2 + S_{xx} \right) \\
&= \frac{1}{nS_{xx}} \left(n(x_i^2 - 2x_i\bar{x} + \bar{x}^2) + S_{xx} \right) \\
&= \frac{1}{nS_{xx}} \left(n(x_i - \bar{x})^2 + S_{xx} \right) \\
&= \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.
\end{aligned}$$

□

19 Lecture 19: Cook's Distance

19.1 Outliers vs. Influential Points

How do observations have a influence on the fitted regression equation? We can detect influential points by the following.

Theorem 19.1. *We say that the ith observation is influential if its exclusion may cause major changes in the fitted regression coefficients. i.e., when $\hat{\beta} - \hat{\beta}_{-i}$ is large, then $\hat{\beta}_{-i}$ is the estimate regression coefficient when the ith observation is not included. We can use Cook's Distance to determine this.*

$$D_i = \frac{(\hat{\beta} - \hat{\beta}_{(-i)})^\top \left(\hat{\sigma}^2 (X^\top X)^{-1}\right)^{-1} (\hat{\beta} - \hat{\beta}_{(-i)})}{p+1} = \frac{(\hat{\beta} - \hat{\beta}_{(-i)})^\top X^\top X (\hat{\beta} - \hat{\beta}_{(-i)})}{\hat{\sigma}^2 (p+1)}.$$

More specifically, we can show that when $D_i = \dfrac{h_{ii} d_i^2}{(1-h_{ii})(p+1)}$ where $d_i = \dfrac{r_i}{\sqrt{\hat{\sigma}^2(1-h_{ii})}}$ is the studentized residual.

Proof. let $x_{(i)}$ be X without row i,

$$x = \begin{pmatrix} x_i \\ x_{(i)} \end{pmatrix}, \qquad y = \begin{pmatrix} y_i \\ y_{(i)} \end{pmatrix}.$$

so

$$y_{(i)} = X_{(i)}\beta + \varepsilon_{(i)}, \qquad \hat{\beta}_{(i)} = (X_{(i)}^\top X_{(i)})^{-1} X_{(i)}^\top y_{(i)}.$$
$$X^\top X \approx X_{(i)}^\top X_{(i)} + x_i x_i^\top, \qquad X^\top y \approx X_{(i)}^\top y_{(i)} + x_i y_i.$$

Then

$$\begin{aligned}
\hat{\beta}_{(i)} &= (X^\top X - x_i x_i^\top)^{-1}(X^\top y - x_i y_i) \\
&= \left[(X^\top X)^{-1} + \frac{(X^\top X)^{-1} x_i x_i^\top (X^\top X)^{-1}}{1 - x_i^\top (X^\top X)^{-1} x_i}\right]\left[X^\top y - x_i y_i\right] \\
&= (X^\top X)^{-1} X^\top y - (X^\top X)^{-1} x_i y_i + \frac{(X^\top X)^{-1} x_i x_i^\top (X^\top X)^{-1}}{1 - h_{ii}} X^\top y - \frac{(X^\top X)^{-1} x_i x_i^\top (X^\top X)^{-1}}{1 - h_{ii}} x_i y_i \\
&= \hat{\beta} - (X^\top X)^{-1} x_i y_i + \frac{(X^\top X)^{-1} x_i \, x_i^\top \hat{\beta}}{1 - h_{ii}} - \frac{(X^\top X)^{-1} x_i \, h_{ii} y_i}{1 - h_{ii}} \\
&= \hat{\beta} - (X^\top X)^{-1} x_i \frac{1}{1 - h_{ii}} \left[y_i - x_i^\top \hat{\beta}\right] \\
&= \hat{\beta} - (X^\top X)^{-1} x_i \frac{r_i}{1 - h_{ii}}.
\end{aligned}$$

$$\implies \hat{\beta} - \hat{\beta}_{(i)} = \frac{r_i}{1 - h_{ii}} (X^\top X)^{-1} x_i.$$

$$(\hat{\beta} - \hat{\beta}_{(i)})^\top X^\top X (\hat{\beta} - \hat{\beta}_{(i)}) = \frac{r_i^2}{(1-h_{ii})^2} x_i^\top (X^\top X)^{-1} X^\top X (X^\top X)^{-1} x_i.$$

$$\text{Then } D_i = \frac{(\hat{\beta} - \hat{\beta}_{(i)})^\top X^\top X (\hat{\beta} - \hat{\beta}_{(i)})}{\hat{\sigma}^2 (p+1)} = \frac{d_i^2 \, x_i^\top (X^\top X)^{-1} x_i}{(1-h_{ii})(p+1)}, \qquad d_i = \frac{r_i}{\hat{\sigma}\sqrt{1-h_{ii}}}.$$

□

Remark 19.1. This is an overall measure i.e., if at least one of the h_{ii}'s or $|d_i|$ is small, then the influence will be small. A large value implies that the observation has a large influence on the results. Cook suggests D_i is significantly large when it is greater than $F_{0.5}(p+1, N-p-1)$ but many data analysts just check if $D_i > 1$ or is obviously greater than others.

Proof.

$$\begin{aligned}\text{Press residual } r_{(c,-i)} &= y_i - x_i^\top \hat{\beta}_{(c,-i)} \\ &= y_i - x_i^\top \hat{\beta} + x_i^\top (\hat{\beta} - \hat{\beta}_{(i)}) \\ &= r_i + x_i^\top \frac{r_i}{1-h_{ii}} (X^\top X)^{-1} x_i \\ &= r_i + \frac{r_i h_{ii}}{1-h_{ii}} \\ &= \frac{r_i}{1-h_{ii}}.\end{aligned}$$

Prediction Error Sum of Squares (PRESS) $\displaystyle\sum_{i=1}^N r_{(c,-i)}^2 = \sum_{i=1}^N \frac{r_i^2}{(1-h_{ii})^2}.$

Difference in Fitted Value (DEFITS)
$$\begin{aligned}\hat{y}_i - \hat{y}_{c,(-i)} &= \hat{y}_i - x_i^\top \hat{\beta}_{(i)} \\ &= (y_i - x_i^\top \hat{\beta}_{(i)}) - (y_i - \hat{y}_i) \\ &= r_{(c,-i)} - r_i \\ &= \frac{r_i}{1-h_{ii}} - r_i \\ &= \frac{h_{ii}}{1-h_{ii}} r_i.\end{aligned}$$

Standardized DEFITS: $\displaystyle \text{DEFITS}_i = \frac{\hat{y}_i - \hat{y}_{c,(-i)}}{\sqrt{\hat{\sigma}^2_{(-i)} h_{ii}}} = \frac{\frac{h_{ii}}{1-h_{ii}} r_i}{\sqrt{\hat{\sigma}^2_{(-i)} h_{ii}}} = \frac{\sqrt{h_{ii}}\, r_i}{(1-h_{ii})\sqrt{\hat{\sigma}^2_{(-i)}}}.$

$\displaystyle \hat{\sigma}^2_{(-i)} = \frac{\sum_{j \neq i} (y_j - x_j^\top \hat{\beta}_{(-i)})^2}{n-p-2}$ is unbiased estimator of σ^2 w/o ith obs.

$$D_i = \frac{h_{ii} d_i^2}{(1-h_{ii})(p+1)} \quad \Rightarrow \quad D_i(p+1) = \frac{h_{ii} d_i^2}{1-h_{ii}} \quad \Rightarrow \quad \sqrt{D_i(p+1)} = \frac{\sqrt{h_{ii}}\, d_i}{1-h_{ii}}.$$

$$d_i = \frac{r_i}{\hat{\sigma}\sqrt{1-h_{ii}}} \quad \Rightarrow \quad \sqrt{D_i(p+1)} = \frac{\sqrt{h_{ii}}\, r_i}{(1-h_{ii})\hat{\sigma}} = \text{DEFITS}_i.$$

□

20 Lecture 21: Singular Value Decomposition of Least Squares

In this lecture we rewrite ordinary least squares in the language of the singular value decomposition (SVD). This makes the roles of multicollinearity, ill-conditioning, and principal components completely transparent. Throughout, let

$$Y \in \mathbb{R}^N, \qquad X \in \mathbb{R}^{N \times p},$$

and consider the usual linear model

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim (0, \sigma^2 I_N).$$

20.1 Review of OLS and Normal Equations

Recall that the ordinary least squares (OLS) estimator $\hat{\beta}$ is defined as

$$\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^p} \ \|Y - X\beta\|_2^2.$$

Define the objective

$$J(\beta) = \|Y - X\beta\|_2^2.$$

We expand $J(\beta)$ explicitly:

$$\begin{aligned} J(\beta) &= (Y - X\beta)^\top (Y - X\beta) \\ &= Y^\top Y - Y^\top X\beta - \beta^\top X^\top Y + \beta^\top X^\top X\beta. \end{aligned}$$

Since $Y^\top X\beta$ is a scalar, it equals its transpose:

$$Y^\top X\beta = (Y^\top X\beta)^\top = \beta^\top X^\top Y.$$

Hence

$$J(\beta) = Y^\top Y - 2\,\beta^\top X^\top Y + \beta^\top X^\top X\beta.$$

To find the minimizer, we differentiate $J(\beta)$ with respect to β and set the gradient to zero. Using the standard matrix calculus rules

$$\frac{\partial}{\partial \beta}\left(\beta^\top A\beta\right) = (A + A^\top)\beta \quad \text{and} \quad \frac{\partial}{\partial \beta}\left(b^\top \beta\right) = b,$$

and the fact that $X^\top X$ is symmetric, we obtain

$$\nabla_\beta J(\beta) = -2X^\top Y + 2X^\top X\beta.$$

Setting the gradient to zero gives the *normal equations*:

$$\begin{aligned} \nabla_\beta J(\beta) = 0 \quad &\iff \quad -2X^\top Y + 2X^\top X\hat{\beta} = 0 \\ &\iff \quad X^\top X\hat{\beta} = X^\top Y. \end{aligned}$$

If $X^\top X$ is invertible (equivalently, if $\operatorname{rank}(X) = p$, i.e., the columns of X are linearly independent), then the OLS solution is unique and given by

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

It is often convenient to introduce the *hat matrix*

$$H = X(X^\top X)^{-1} X^\top,$$

so that

$$\hat{Y} = X\hat{\beta} = X(X^\top X)^{-1} X^\top Y = HY$$

is the orthogonal projection of Y onto the column space $C(X)$.

Remark 20.1. Geometrically, OLS finds the point $\hat{Y}$ in the p-dimensional subspace $C(X) \subset \mathbb{R}^N$ that is closest to Y in Euclidean distance. The residual vector $r = Y - \hat{Y}$ lies in $C(X)^\perp$, the orthogonal complement of $C(X)$.

20.2 Singular Value Decomposition of X

We now introduce the singular value decomposition (SVD) of the design matrix X.

Definition 20.1 (Singular value decomposition)**.** Let $X \in \mathbb{R}^{N\times p}$ have rank $r = \operatorname{rank}(X)$. Then there exist orthogonal matrices

$$U \in \mathbb{R}^{N\times N}, \qquad V \in \mathbb{R}^{p\times p},$$

and a diagonal matrix

$$\Sigma \in \mathbb{R}^{N\times p}$$

such that

$$X = U\Sigma V^\top.$$

The diagonal entries of Σ are the singular values $\sigma_1, \dots, \sigma_r > 0$, arranged in non-increasing order,

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,$$

with all remaining diagonal entries equal to 0. The columns $u_1, \dots, u_N$ of U are the left singular vectors, and the columns $v_1, \dots, v_p$ of V are the right singular vectors.

Explicitly, we can write the SVD as

$$X = \sum_{j=1}^{r} \sigma_j u_j v_j^\top,$$

so X is a sum of r rank-one matrices.

20.2.1 Thin vs. Full SVD

For least squares, it is convenient to work with the *thin SVD*. Let $r = \operatorname{rank}(X)$. Define

$$U_r = [u_1, \dots, u_r] \in \mathbb{R}^{N\times r}, \quad V_r = [v_1, \dots, v_r] \in \mathbb{R}^{p\times r}, \quad \Sigma_r = \operatorname{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}^{r\times r}.$$

Then we have the *thin SVD*

$$X = U_r \Sigma_r V_r^\top,$$

where $U_r^\top U_r = I_r$ and $V_r^\top V_r = I_r$. The full SVD uses $U \in \mathbb{R}^{N\times N}$ and $V \in \mathbb{R}^{p\times p}$, with

$$\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix},$$

but the thin SVD keeps only the r nonzero singular values and their singular vectors.

Remark 20.2. If X has full column rank ($r = p$ and $p \leq N$), then the thin SVD has $U_p \in \mathbb{R}^{N\times p}$, $\Sigma_p \in \mathbb{R}^{p\times p}$, $V_p \in \mathbb{R}^{p\times p}$, and we may simply write

$$X = U\Sigma V^\top,$$

understanding that Σ is square and invertible.

20.2.2 Rank, singular values, and condition number

The singular values of X immediately encode its rank and well-conditionedness.

- The rank is
$$r = \#\{j : \sigma_j > 0\}.$$
- The eigenvalues of $X^\top X$ are $\lambda_j = \sigma_j^2$ for $j = 1, \dots, r$, with the remaining $p - r$ eigenvalues equal to 0.

The *spectral condition number* of X (restricted to its column space) is

$$\kappa(X) = \frac{\sigma_1}{\sigma_r} \quad (\text{assuming } r \geq 1),$$

and the condition number of the Gram matrix $X^\top X$ is

$$\kappa(X^\top X) = \frac{\lambda_{\max}(X^\top X)}{\lambda^+_{\min}(X^\top X)} = \frac{\sigma_1^2}{\sigma_r^2} = \kappa(X)^2,$$

where $\lambda^+_{\min}(X^\top X)$ denotes the smallest positive eigenvalue. If some singular values are very small (though nonzero), $X^\top X$ is ill-conditioned and the OLS estimator becomes numerically unstable.

20.3 OLS via SVD

We now express the OLS solution entirely in terms of the SVD of X. For clarity, first assume full column rank: $r = p$ and $p \leq N$, so

$$X = U\Sigma V^\top, \quad \Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_p), \quad \sigma_j > 0.$$

20.3.1 Moore–Penrose pseudoinverse X^+

The SVD gives a very simple expression for the *Moore–Penrose pseudoinverse* of X.

Definition 20.2 (Pseudoinverse via SVD)**.** Let $X = U\Sigma V^\top$ be the SVD of X. Define

$$\Sigma^+ \in \mathbb{R}^{p \times N}$$

by taking the transpose shape of Σ and replacing each nonzero singular value σ_j with its reciprocal $1/\sigma_j$, and leaving zeros as zeros:

$$\Sigma^+ = \operatorname{diag}(1/\sigma_1, \dots, 1/\sigma_r) \quad (\text{embedded in a } p \times N \text{ matrix}).$$

Then the Moore–Penrose pseudoinverse of X is

$$X^+ = V\Sigma^+ U^\top.$$

In the full column rank case ($r = p$), the pseudoinverse reduces to the usual inverse formula:

$$X^+ = V\Sigma^{-1}U^\top,$$

and we can check that this equals $(X^\top X)^{-1}X^\top$. Indeed,

$$\begin{aligned} X^\top X &= (U\Sigma V^\top)^\top (U\Sigma V^\top) \\ &= V\Sigma^\top U^\top U\Sigma V^\top \\ &= V\Sigma^\top \Sigma V^\top \\ &= V\Sigma^2 V^\top, \end{aligned}$$

since Σ is diagonal with positive entries and $\Sigma^\top = \Sigma$. Thus

$$\begin{aligned}(X^\top X)^{-1} &= \left(V\Sigma^2 V^\top\right)^{-1} \\ &= V(\Sigma^2)^{-1}V^\top \\ &= V\Sigma^{-2}V^\top.\end{aligned}$$

Hence

$$\begin{aligned}(X^\top X)^{-1}X^\top &= V\Sigma^{-2}V^\top(U\Sigma V^\top)^\top \\ &= V\Sigma^{-2}V^\top V\Sigma U^\top \\ &= V\Sigma^{-1}U^\top \\ &= X^+.\end{aligned}$$

So in the full column rank case, the pseudoinverse agrees with the usual OLS matrix $(X^\top X)^{-1}X^\top$.

Remark 20.3. In the rank-deficient case ($r < p$), $X^\top X$ is not invertible. However, $X^+ = V\Sigma^+U^\top$ always exists and gives the *minimum-norm* least squares solution

$$\hat{\beta} = X^+Y.$$

This is the solution with smallest $\|\beta\|_2$ among all vectors β that minimize $\|Y - X\beta\|_2$.

20.3.2 Expression for $\hat{\beta}$ in terms of U, Σ, V

Derivation via normal equations. Assume full column rank and write $X = U\Sigma V^\top$ as above. The normal equations are

$$X^\top X\hat{\beta} = X^\top Y.$$

Using the SVD,

$$\begin{aligned}X^\top X &= V\Sigma^2 V^\top, \\ X^\top Y &= (U\Sigma V^\top)^\top Y = V\Sigma U^\top Y.\end{aligned}$$

Thus

$$V\Sigma^2 V^\top\hat{\beta} = V\Sigma U^\top Y.$$

Left-multiply both sides by $V^\top$ to move into the eigenbasis:

$$\Sigma^2 V^\top\hat{\beta} = \Sigma U^\top Y.$$

Since Σ is invertible (full column rank case), we may multiply by Σ^{-1} on the left:

$$\begin{aligned}\Sigma^{-1}\Sigma^2 V^\top\hat{\beta} &= \Sigma^{-1}\Sigma U^\top Y \\ \Sigma V^\top\hat{\beta} &= U^\top Y.\end{aligned}$$

Now multiply by Σ^{-1} again:

$$V^\top\hat{\beta} = \Sigma^{-1}U^\top Y.$$

Finally, multiply by V:

$$\hat{\beta} = V\Sigma^{-1}U^\top Y.$$

Using the pseudoinverse notation, this is simply

$$\hat{\beta} = X^+Y.$$

Derivation by orthogonal change of coordinates. We can also derive $\hat{\beta}$ by minimizing the loss in a rotated coordinate system. Using the SVD,

$$J(\beta) = \|Y - X\beta\|_2^2 = \|Y - U\Sigma V^\top \beta\|_2^2.$$

Because U is orthogonal, $\|y\|_2 = \|U^\top y\|_2$ for any y, so

$$\begin{aligned} J(\beta) &= \|U^\top Y - U^\top U\Sigma V^\top \beta\|_2^2 \\ &= \|\tilde{Y} - \Sigma\tilde{\beta}\|_2^2, \end{aligned}$$

where we have defined the rotated variables

$$\tilde{Y} = U^\top Y \in \mathbb{R}^N, \qquad \tilde{\beta} = V^\top \beta \in \mathbb{R}^p.$$

Write $\tilde{Y} = (\tilde{Y}_1, \ldots, \tilde{Y}_N)^\top$ and $\tilde{\beta} = (\tilde{\beta}_1, \ldots, \tilde{\beta}_p)^\top$. Since Σ is diagonal with $\sigma_1, \ldots, \sigma_p$ on the diagonal,

$$\Sigma\tilde{\beta} = (\sigma_1\tilde{\beta}_1, \ldots, \sigma_p\tilde{\beta}_p, 0, \ldots, 0)^\top.$$

Hence

$$J(\beta) = \sum_{j=1}^{p} (\tilde{Y}_j - \sigma_j\tilde{\beta}_j)^2 + \sum_{j=p+1}^{N} \tilde{Y}_j^2.$$

The second sum does not depend on $\tilde{\beta}$, so minimizing J is equivalent to minimizing the first sum. Each coordinate $\tilde{\beta}_j$ appears only in one term

$$(\tilde{Y}_j - \sigma_j\tilde{\beta}_j)^2,$$

so the minimizer is obtained by solving p scalar least squares problems:

$$\frac{\partial}{\partial\tilde{\beta}_j}(\tilde{Y}_j - \sigma_j\tilde{\beta}_j)^2 = -2\sigma_j(\tilde{Y}_j - \sigma_j\tilde{\beta}_j) = 0,$$

which gives

$$\tilde{\beta}_j = \frac{\tilde{Y}_j}{\sigma_j}, \qquad j = 1, \ldots, p.$$

Thus

$$\tilde{\beta} = \Sigma^{-1}\tilde{Y} = \Sigma^{-1}U^\top Y.$$

Returning to the original coordinates,

$$\hat{\beta} = V\tilde{\beta} = V\Sigma^{-1}U^\top Y.$$

Again, this is the same X^+Y as before.

20.4 Multicollinearity in the SVD Basis

The SVD gives a clean description of how multicollinearity leads to large variance in the OLS estimator. Recall that in the full column rank case

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Write the model as

$$Y = X\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon] = 0, \qquad \mathrm{Var}(\varepsilon) = \sigma^2 I_N.$$

Then

$$\begin{aligned}\hat{\beta} &= (X^\top X)^{-1} X^\top (X\beta + \varepsilon) \\ &= (X^\top X)^{-1} X^\top X\beta + (X^\top X)^{-1} X^\top \varepsilon \\ &= \beta + (X^\top X)^{-1} X^\top \varepsilon.\end{aligned}$$

Thus

$$\mathbb{E}[\hat{\beta}] = \beta \quad \text{and} \quad \mathrm{Var}(\hat{\beta}) = \mathrm{Var}\big((X^\top X)^{-1} X^\top \varepsilon\big) = \sigma^2 (X^\top X)^{-1}.$$

20.4.1 Small singular values and variance explosion

Using the SVD, we can express $(X^\top X)^{-1}$ in terms of singular values and singular vectors. From $X = U\Sigma V^\top$,

$$X^\top X = V\Sigma^2 V^\top,$$

so

$$(X^\top X)^{-1} = V\Sigma^{-2} V^\top.$$

Therefore

$$\mathrm{Var}(\hat{\beta}) = \sigma^2 V\Sigma^{-2} V^\top.$$

It is instructive to examine $\hat{\beta}$ in the rotated basis of right singular vectors. Define

$$\tilde{\beta} = V^\top \beta, \qquad \tilde{\hat{\beta}} = V^\top \hat{\beta}.$$

Then

$$\tilde{\hat{\beta}} = V^\top \hat{\beta} = V^\top \beta + V^\top (X^\top X)^{-1} X^\top \varepsilon.$$

The mean is

$$\mathbb{E}[\tilde{\hat{\beta}}] = \tilde{\beta},$$

and the covariance is

$$\begin{aligned}\mathrm{Var}(\tilde{\hat{\beta}}) &= V^\top \mathrm{Var}(\hat{\beta}) V \\ &= V^\top \big(\sigma^2 V\Sigma^{-2} V^\top\big) V \\ &= \sigma^2 \Sigma^{-2},\end{aligned}$$

since $V^\top V = I_p$. Because Σ^{-2} is diagonal with entries $1/\sigma_j^2$, this means

$$\mathrm{Var}(\tilde{\hat{\beta}}_j) = \frac{\sigma^2}{\sigma_j^2}, \qquad j = 1, \dots, p.$$

- Directions (right singular vectors) v_j corresponding to *large* singular values σ_j have *small* variance σ^2/σ_j^2.
- Directions corresponding to *small* singular values σ_j have *huge* variance σ^2/σ_j^2.

Thus, if some σ_j are very close to zero (near linear dependence among columns of X), the variance of the OLS estimator in those directions explodes. This is precisely the phenomenon of *multicollinearity*: small perturbations in Y cause enormous changes in the estimated coefficients along nearly collinear directions.

20.4.2 Geometric picture: elongated ellipsoids

We can also see multicollinearity geometrically. Consider the loss

$$J(\beta) = \|Y - X\beta\|_2^2.$$

Near the minimum $\hat{\beta}$, we can approximate $J(\beta)$ by a quadratic form. Expanding around $\hat{\beta}$,

$$\Delta\beta := \beta - \hat{\beta},$$

we have (up to an irrelevant constant)

$$J(\beta) - J(\hat{\beta}) \approx \Delta\beta^\top X^\top X \Delta\beta.$$

Using the SVD,

$$X^\top X = V\Sigma^2 V^\top,$$

so

$$\Delta\beta^\top X^\top X \Delta\beta = \Delta\beta^\top V\Sigma^2 V^\top \Delta\beta.$$

Let

$$\gamma = V^\top \Delta\beta,$$

so $\Delta\beta = V\gamma$. Then

$$\Delta\beta^\top X^\top X \Delta\beta = \gamma^\top \Sigma^2 \gamma = \sum_{j=1}^{p} \sigma_j^2 \gamma_j^2.$$

Fix a constant $c > 0$ and consider the level set

$$\{\beta : J(\beta) - J(\hat{\beta}) = c\}.$$

In the coordinates γ_j, this becomes

$$\sum_{j=1}^{p} \sigma_j^2 \gamma_j^2 = c,$$

which is an ellipsoid with semi-axis lengths

$$\sqrt{\frac{c}{\sigma_1^2}}, \ldots, \sqrt{\frac{c}{\sigma_p^2}}.$$

- If σ_j is large, the ellipsoid is *short* in the v_j direction: moving a small amount in that direction increases the loss significantly.
- If σ_j is very small, the ellipsoid is *very elongated* in the v_j direction: the loss changes very slowly as we move along that direction.

Thus, in the presence of multicollinearity (small σ_j), the loss surface is almost flat along certain directions v_j, and the least-squares solution is highly unstable in those directions. Noise in Y can push the estimator a long way along these nearly-flat directions without changing the fit much, causing large variance in the coefficients.

20.5 Principal Component Regression (PCR) – Preview

Principal Component Regression (PCR) uses the SVD geometry to construct a more stable regression estimator by restricting attention to the *principal directions* associated with large singular values.

20.5.1 Rotating to the principal component basis

From the SVD,

$$X = U\Sigma V^\top.$$

The columns of V are eigenvectors of $X^\top X$, and the columns of U are eigenvectors of $XX^\top$. The matrix of *principal component scores* is

$$Z := XV = U\Sigma V^\top V = U\Sigma.$$

The j-th column of Z is

$$z_j = \sigma_j u_j,$$

the j-th principal component of the design. We can reparameterize the regression model in terms of principal components. Let

$$\gamma = V^\top \beta \quad \Longleftrightarrow \quad \beta = V\gamma.$$

Then

$$\begin{aligned} Y &= X\beta + \varepsilon \\ &= XV\gamma + \varepsilon \\ &= Z\gamma + \varepsilon \\ &= U\Sigma\gamma + \varepsilon. \end{aligned}$$

So we are regressing Y on the principal components (columns of Z) with coefficients γ. The OLS estimator for γ in this model is

$$\hat{\gamma} = (Z^\top Z)^{-1} Z^\top Y,$$

provided $Z^\top Z$ is invertible. Using $Z = XV$ and $X^\top X = V\Sigma^2 V^\top$, we compute:

$$\begin{aligned} Z^\top Z &= V^\top X^\top XV \\ &= V^\top (V\Sigma^2 V^\top) V \\ &= \Sigma^2, \\ Z^\top Y &= V^\top X^\top Y = V^\top X^\top Y. \end{aligned}$$

But from before, we know

$$X^\top Y = V\Sigma U^\top Y,$$

so

$$Z^\top Y = V^\top (V\Sigma U^\top Y) = \Sigma U^\top Y.$$

Thus

$$\begin{aligned} \hat{\gamma} &= (Z^\top Z)^{-1} Z^\top Y \\ &= \Sigma^{-2}\Sigma U^\top Y \\ &= \Sigma^{-1} U^\top Y. \end{aligned}$$

The corresponding coefficient vector in the original basis is

$$\hat{\beta} = V\hat{\gamma} = V\Sigma^{-1}U^\top Y = X^+ Y,$$

exactly the same OLS solution as before. Thus, OLS can be viewed as:

1. Rotate the predictors to the principal component basis (V),
2. Regress Y on all principal components,
3. Rotate back to the original coordinates.

20.5.2 Truncating small singular values as implicit regularization

Principal Component Regression modifies this procedure by discarding (or shrinking) directions associated with small singular values. Let $m < r$ and define

$$V_m = [v_1, \dots, v_m] \in \mathbb{R}^{p\times m}, \quad U_m = [u_1, \dots, u_m] \in \mathbb{R}^{N\times m}, \quad \Sigma_m = \operatorname{diag}(\sigma_1, \dots, \sigma_m).$$

The first m principal components are

$$Z_m := XV_m = U_m\Sigma_m.$$

PCR with m components fits the reduced model

$$Y = Z_m\gamma_m + \varepsilon,$$

with $\gamma_m \in \mathbb{R}^m$. The OLS estimator is

$$\hat{\gamma}_m = (Z_m^\top Z_m)^{-1} Z_m^\top Y.$$

Compute

$$Z_m^\top Z_m = \Sigma_m U_m^\top U_m \Sigma_m = \Sigma_m^2,$$
$$Z_m^\top Y = \Sigma_m U_m^\top Y.$$

Thus

$$\hat{\gamma}_m = \Sigma_m^{-2}\Sigma_m U_m^\top Y = \Sigma_m^{-1} U_m^\top Y.$$

The corresponding PCR estimator in the original coefficient space is

$$\hat{\beta}_{\mathrm{PCR},m} = V_m\hat{\gamma}_m = V_m\Sigma_m^{-1}U_m^\top Y.$$

This can be viewed as applying a *truncated pseudoinverse*:

$$X^+_{(m)} := V_m\Sigma_m^{-1}U_m^\top, \qquad \hat{\beta}_{\mathrm{PCR},m} = X^+_{(m)}Y.$$

Remark 20.4 (Implicit regularization). Comparing OLS and PCR:

$$\hat{\beta}_{\mathrm{OLS}} = \sum_{j=1}^{r} \frac{1}{\sigma_j} v_j u_j^\top Y,$$
$$\hat{\beta}_{\mathrm{PCR},m} = \sum_{j=1}^{m} \frac{1}{\sigma_j} v_j u_j^\top Y.$$

PCR keeps only the directions v_j with *large* singular values σ_j and completely discards directions with small σ_j. Instead of dividing by tiny σ_j and exploding the variance of coefficients, PCR removes those unstable directions from the model entirely. This is a form of *regularization* in the SVD basis.

21 Lecture 22: Ridge Regression

21.1 Motivation for Shrinkage

We work with the usual linear model

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim (0, \sigma^2 I_N),$$

with $X \in \mathbb{R}^{N\times p}$, $\beta \in \mathbb{R}^p$, $Y \in \mathbb{R}^N$.

21.1.1 Variance of $\hat{\beta}$ under multicollinearity

Recall the OLS estimator

$$\hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top Y$$

when $X^\top X$ is invertible. Write $Y = X\beta + \varepsilon$ and plug into $\hat{\beta}_{\text{OLS}}$:

$$\begin{aligned}\hat{\beta}_{\text{OLS}} &= (X^\top X)^{-1} X^\top (X\beta + \varepsilon) \\ &= (X^\top X)^{-1} X^\top X\beta + (X^\top X)^{-1} X^\top \varepsilon \\ &= \beta + (X^\top X)^{-1} X^\top \varepsilon.\end{aligned}$$

Hence

$$\mathbb{E}[\hat{\beta}_{\text{OLS}}] = \beta$$

and

$$\begin{aligned}\text{Var}(\hat{\beta}_{\text{OLS}}) &= \text{Var}\left((X^\top X)^{-1} X^\top \varepsilon\right) \\ &= (X^\top X)^{-1} X^\top \text{Var}(\varepsilon) X (X^\top X)^{-1} \\ &= (X^\top X)^{-1} X^\top (\sigma^2 I_N) X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} \\ &= \sigma^2 (X^\top X)^{-1}.\end{aligned}$$

Diagonalize $X^\top X$ via an eigen-decomposition

$$X^\top X = V \Lambda V^\top,$$

with $V \in \mathbb{R}^{p\times p}$ orthogonal and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p > 0$. Then

$$\begin{aligned}(X^\top X)^{-1} &= (V \Lambda V^\top)^{-1} \\ &= V \Lambda^{-1} V^\top,\end{aligned}$$

so

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = \sigma^2 V \Lambda^{-1} V^\top.$$

In the rotated coordinates $\tilde{\beta}_{\text{OLS}} = V^\top \hat{\beta}_{\text{OLS}}$,

$$\begin{aligned}\text{Var}(\tilde{\beta}_{\text{OLS}}) &= V^\top \text{Var}(\hat{\beta}_{\text{OLS}}) V \\ &= V^\top (\sigma^2 V \Lambda^{-1} V^\top) V \\ &= \sigma^2 \Lambda^{-1}.\end{aligned}$$

Thus

$$\text{Var}(\tilde{\beta}_{\text{OLS},j}) = \frac{\sigma^2}{\lambda_j}.$$

When X is nearly rank-deficient (multicollinearity), some λ_j are very small, so σ^2/λ_j is very large, and the OLS coefficients have huge variance in those directions.

21.1.2 Bias–variance tradeoff in linear models

Consider a linear estimator of the form

$$\tilde{\beta} = AY,$$

for some fixed matrix $A \in \mathbb{R}^{p\times N}$. Using $Y = X\beta + \varepsilon$, we have

$$\begin{aligned}\tilde{\beta} &= A(X\beta + \varepsilon)\\ &= AX\beta + A\varepsilon.\end{aligned}$$

Then

$$\begin{aligned}\mathbb{E}[\tilde{\beta}] &= AX\beta,\\ \operatorname{Bias}(\tilde{\beta}) &= \mathbb{E}[\tilde{\beta}] - \beta = (AX - I_p)\beta,\\ \operatorname{Var}(\tilde{\beta}) &= A\operatorname{Var}(\varepsilon)A^\top = A(\sigma^2 I_N)A^\top = \sigma^2 AA^\top.\end{aligned}$$

For the mean squared error matrix

$$\operatorname{MSE}(\tilde{\beta}) = \mathbb{E}\big[(\tilde{\beta}-\beta)(\tilde{\beta}-\beta)^\top\big] = \operatorname{Bias}(\tilde{\beta})\operatorname{Bias}(\tilde{\beta})^\top + \operatorname{Var}(\tilde{\beta}),$$

we see that we can reduce variance (via A) at the cost of introducing bias $(AX - I_p)\beta$. Ridge chooses A to deliberately shrink coefficients toward zero (introducing bias) but reduce variance substantially in collinear directions.

21.2 Ridge Regression: Penalized Least Squares

21.2.1 Constrained vs. penalized formulation

The constrained form of ridge is

$$\min_{\beta\in\mathbb{R}^p} \|Y - X\beta\|_2^2 \quad \text{subject to} \quad \|\beta\|_2^2 \le t,$$

for some radius $t > 0$. Form the Lagrangian

$$\mathcal{L}(\beta,\mu) = \|Y - X\beta\|_2^2 + \mu\big(\|\beta\|_2^2 - t\big), \quad \mu \ge 0.$$

For any fixed μ, the term $-\mu t$ is constant in β. Define $\lambda := \mu$ and the penalized objective

$$J_\lambda(\beta) = \|Y - X\beta\|_2^2 + \lambda\|\beta\|_2^2, \quad \lambda \ge 0.$$

For each t there is a corresponding λ (and vice versa) such that the solution of the constrained problem equals the minimizer of J_λ. Thus ridge can be seen equivalently as a quadratic constraint or a quadratic penalty.

21.2.2 Normal equations: $(X^\top X + \lambda I)\hat{\beta}_\lambda = X^\top Y$

Define

$$\hat{\beta}_\lambda = \arg\min_{\beta \in \mathbb{R}^p} J_\lambda(\beta) = \arg\min_{\beta \in \mathbb{R}^p} \left\{ \|Y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}, \quad \lambda \geq 0.$$

Expand $J_\lambda(\beta)$:

$$\begin{aligned} J_\lambda(\beta) &= (Y - X\beta)^\top (Y - X\beta) + \lambda \beta^\top \beta \\ &= Y^\top Y - Y^\top X\beta - \beta^\top X^\top Y + \beta^\top X^\top X\beta + \lambda \beta^\top \beta. \end{aligned}$$

Since $Y^\top X\beta$ is scalar,

$$Y^\top X\beta = \beta^\top X^\top Y,$$

so

$$\begin{aligned} J_\lambda(\beta) &= Y^\top Y - 2\beta^\top X^\top Y + \beta^\top X^\top X\beta + \lambda \beta^\top \beta \\ &= Y^\top Y - 2\beta^\top X^\top Y + \beta^\top (X^\top X + \lambda I_p)\beta. \end{aligned}$$

Differentiate with respect to β. Using symmetry of $X^\top X + \lambda I_p$ and the same rules as before:

$$\nabla_\beta J_\lambda(\beta) = -2X^\top Y + 2(X^\top X + \lambda I_p)\beta.$$

Set the gradient to zero:

$$\begin{aligned} \nabla_\beta J_\lambda(\hat{\beta}_\lambda) = 0 &\iff -2X^\top Y + 2(X^\top X + \lambda I_p)\hat{\beta}_\lambda = 0 \\ &\iff (X^\top X + \lambda I_p)\hat{\beta}_\lambda = X^\top Y. \end{aligned}$$

When $\lambda > 0$, the matrix $X^\top X + \lambda I_p$ is strictly positive definite and invertible, even if $X^\top X$ is singular. Thus

$$\hat{\beta}_\lambda = (X^\top X + \lambda I_p)^{-1} X^\top Y.$$

21.3 Ridge via SVD / Eigen-Decomposition

21.3.1 Expression in SVD basis: shrinkage of $1/\sigma_j$

Use the SVD of X as in Lecture 20:

$$X = U\Sigma V^\top,$$

where $U \in \mathbb{R}^{N \times N}$, $V \in \mathbb{R}^{p \times p}$ orthogonal, and

$$\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_p), \quad \sigma_1 \geq \cdots \geq \sigma_p \geq 0.$$

Assume full column rank so $\sigma_j > 0$ for $j = 1, \ldots, p$. Compute $X^\top X$ and the ridge matrix:

$$\begin{aligned} X^\top X &= (U\Sigma V^\top)^\top (U\Sigma V^\top) \\ &= V\Sigma^\top U^\top U\Sigma V^\top \\ &= V\Sigma^2 V^\top, \\ X^\top X + \lambda I_p &= V\Sigma^2 V^\top + \lambda I_p \\ &= V\Sigma^2 V^\top + \lambda V I_p V^\top \\ &= V(\Sigma^2 + \lambda I_p)V^\top. \end{aligned}$$

Invert:

$$\begin{aligned} (X^\top X + \lambda I_p)^{-1} &= \left(V(\Sigma^2 + \lambda I_p)V^\top\right)^{-1} \\ &= V(\Sigma^2 + \lambda I_p)^{-1} V^\top, \end{aligned}$$

since V is orthogonal and $(\Sigma^2 + \lambda I_p)$ is diagonal with entries $\sigma_j^2 + \lambda$. Now

$$\begin{aligned}\hat{\beta}_\lambda &= (X^\top X + \lambda I_p)^{-1} X^\top Y \\ &= V(\Sigma^2 + \lambda I_p)^{-1} V^\top (U\Sigma V^\top)^\top Y \\ &= V(\Sigma^2 + \lambda I_p)^{-1} V^\top V \Sigma U^\top Y \\ &= V(\Sigma^2 + \lambda I_p)^{-1} \Sigma U^\top Y.\end{aligned}$$

Since $(\Sigma^2 + \lambda I_p)^{-1}\Sigma$ is diagonal with entries

$$\frac{\sigma_j}{\sigma_j^2 + \lambda},$$

we have

$$\hat{\beta}_\lambda = V \operatorname{diag}\left(\frac{\sigma_1}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p}{\sigma_p^2 + \lambda}\right) U^\top Y.$$

Compare to the OLS solution

$$\hat{\beta}_{\text{OLS}} = V \operatorname{diag}\left(\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_p}\right) U^\top Y.$$

We can write ridge as

$$\begin{aligned}\hat{\beta}_\lambda &= V \operatorname{diag}\left(\frac{\sigma_1}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p}{\sigma_p^2 + \lambda}\right) U^\top Y \\ &= V \operatorname{diag}\left(\frac{\sigma_1^2}{\sigma_1^2 + \lambda} \cdot \frac{1}{\sigma_1}, \ldots, \frac{\sigma_p^2}{\sigma_p^2 + \lambda} \cdot \frac{1}{\sigma_p}\right) U^\top Y \\ &= V \operatorname{diag}\left(\frac{\sigma_1^2}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p^2}{\sigma_p^2 + \lambda}\right) \operatorname{diag}\left(\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_p}\right) U^\top Y \\ &= V \operatorname{diag}\left(\frac{\sigma_1^2}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p^2}{\sigma_p^2 + \lambda}\right) V^\top \hat{\beta}_{\text{OLS}}.\end{aligned}$$

Thus, in the V-basis (right singular vectors), ridge multiplies each OLS component by the shrinkage factor

$$s_j(\lambda) = \frac{\sigma_j^2}{\sigma_j^2 + \lambda} \in (0, 1).$$

Small σ_j^2 (ill-conditioned directions) get shrunk the most.

21.3.2 Effective degrees of freedom $\mathrm{df}(\lambda)$

The ridge fitted values are

$$\hat{Y}_\lambda = X\hat{\beta}_\lambda.$$

Using $\hat{\beta}_\lambda = (X^\top X + \lambda I_p)^{-1} X^\top Y$, we can write

$$\hat{Y}_\lambda = S_\lambda Y, \quad S_\lambda := X(X^\top X + \lambda I_p)^{-1} X^\top.$$

The matrix S_λ is the ridge *hat matrix*. Using the SVD,

$$\begin{aligned}S_\lambda &= U\Sigma V^\top \big(V(\Sigma^2 + \lambda I_p)V^\top\big)^{-1} V\Sigma U^\top \\ &= U\Sigma V^\top V(\Sigma^2 + \lambda I_p)^{-1} V^\top V \Sigma U^\top \\ &= U\Sigma(\Sigma^2 + \lambda I_p)^{-1}\Sigma U^\top.\end{aligned}$$

The middle matrix is diagonal with entries

$$\frac{\sigma_j^2}{\sigma_j^2+\lambda},$$

so

$$S_\lambda = U\,\mathrm{diag}\left(\frac{\sigma_1^2}{\sigma_1^2+\lambda},\dots,\frac{\sigma_p^2}{\sigma_p^2+\lambda},0,\dots,0\right)U^\top.$$

The effective degrees of freedom for ridge is defined as

$$\mathrm{df}(\lambda) = \mathrm{tr}(S_\lambda).$$

Using the above,

$$\begin{aligned}\mathrm{df}(\lambda) &= \mathrm{tr}\left(U\,\mathrm{diag}(\cdots)U^\top\right)\\ &= \mathrm{tr}\left(\,\mathrm{diag}(\cdots)U^\top U\right)\\ &= \mathrm{tr}\left(\,\mathrm{diag}(\cdots)I_N\right)\\ &= \sum_{j=1}^{p}\frac{\sigma_j^2}{\sigma_j^2+\lambda}.\end{aligned}$$

Note that $\mathrm{df}(0) = p$ (OLS) and $\mathrm{df}(\lambda)\downarrow 0$ as $\lambda\to\infty$.

21.4 Bayesian Interpretation of Ridge

21.4.1 Gaussian prior on β

Assume the likelihood

$$Y \mid \beta \sim N(X\beta, \sigma^2 I_N),$$

and place a Gaussian prior on β:

$$\beta \sim N(0, \tau^2 I_p),$$

independent of ε. The (unnormalized) posterior density is proportional to

$$p(\beta \mid Y) \propto p(Y \mid \beta)\, p(\beta).$$

Write the log posterior (up to an additive constant) as

$$\log p(\beta \mid Y) = -\frac{1}{2\sigma^2}\|Y - X\beta\|_2^2 - \frac{1}{2\tau^2}\|\beta\|_2^2 + \text{const}.$$

Equivalently, maximizing $\log p(\beta \mid Y)$ is equivalent to minimizing

$$\frac{1}{\sigma^2}\|Y - X\beta\|_2^2 + \frac{1}{\tau^2}\|\beta\|_2^2.$$

Multiplying by σ^2 (positive, so argmin is unchanged) gives the ridge objective

$$\|Y - X\beta\|_2^2 + \lambda\|\beta\|_2^2, \quad \lambda = \frac{\sigma^2}{\tau^2}.$$

Thus the posterior mode equals the ridge estimator with $\lambda = \sigma^2/\tau^2$.

21.4.2 Posterior mean = ridge estimator

Because the prior and likelihood are Gaussian, the posterior is also Gaussian. We can compute its mean by completing the square. Write the negative log posterior (up to constants):

$$\ell(\beta) = \frac{1}{2\sigma^2}\|Y - X\beta\|_2^2 + \frac{1}{2\tau^2}\|\beta\|_2^2.$$

Expand the first term:

$$\begin{aligned}\frac{1}{2\sigma^2}\|Y - X\beta\|_2^2 &= \frac{1}{2\sigma^2}(Y - X\beta)^\top(Y - X\beta) \\ &= \frac{1}{2\sigma^2}\big(Y^\top Y - 2\beta^\top X^\top Y + \beta^\top X^\top X\beta\big).\end{aligned}$$

The prior term is

$$\frac{1}{2\tau^2}\|\beta\|_2^2 = \frac{1}{2\tau^2}\beta^\top\beta.$$

Collect terms in β:

$$\begin{aligned}\ell(\beta) &= \frac{1}{2\sigma^2}Y^\top Y - \frac{1}{\sigma^2}\beta^\top X^\top Y + \frac{1}{2\sigma^2}\beta^\top X^\top X\beta + \frac{1}{2\tau^2}\beta^\top\beta + \text{const} \\ &= \text{const} - \frac{1}{\sigma^2}\beta^\top X^\top Y + \frac{1}{2}\beta^\top\left(\frac{1}{\sigma^2}X^\top X + \frac{1}{\tau^2}I_p\right)\beta.\end{aligned}$$

Define

$$A = \frac{1}{\sigma^2}X^\top X + \frac{1}{\tau^2}I_p, \qquad b = \frac{1}{\sigma^2}X^\top Y.$$

Then

$$\ell(\beta) = \text{const} - \beta^\top b + \frac{1}{2}\beta^\top A\beta.$$

Complete the square:

$$\begin{aligned}\ell(\beta) &= \text{const} + \frac{1}{2}\left(\beta^\top A\beta - 2\beta^\top b\right) \\ &= \text{const} + \frac{1}{2}\left[(\beta - A^{-1}b)^\top A(\beta - A^{-1}b) - b^\top A^{-1}b\right].\end{aligned}$$

This is the negative log density of a Gaussian with mean $A^{-1}b$ and covariance matrix A^{-1}, so

$$\beta \mid Y \sim N\left(A^{-1}b,\ A^{-1}\right).$$

Compute $A^{-1}b$ explicitly:

$$\begin{aligned}A &= \frac{1}{\sigma^2}X^\top X + \frac{1}{\tau^2}I_p = \frac{1}{\sigma^2}\left(X^\top X + \frac{\sigma^2}{\tau^2}I_p\right), \\ A^{-1} &= \sigma^2\left(X^\top X + \frac{\sigma^2}{\tau^2}I_p\right)^{-1}, \\ b &= \frac{1}{\sigma^2}X^\top Y.\end{aligned}$$

Then

$$A^{-1}b = \sigma^2\left(X^\top X + \frac{\sigma^2}{\tau^2}I_p\right)^{-1}\left(\frac{1}{\sigma^2}X^\top Y\right) = \left(X^\top X + \frac{\sigma^2}{\tau^2}I_p\right)^{-1}X^\top Y.$$

Thus

$$\mathbb{E}[\beta \mid Y] = \left(X^\top X + \lambda I_p\right)^{-1} X^\top Y, \quad \lambda = \frac{\sigma^2}{\tau^2}.$$

This is exactly the ridge estimator $\hat{\beta}_\lambda$ with $\lambda = \sigma^2/\tau^2$. For Gaussian models, the posterior mean and posterior mode coincide, so ridge is both the MAP and posterior mean.

21.5 Choosing λ

21.5.1 Cross-validation for ridge

Because ridge is a linear smoother

$$\hat{Y}_\lambda = S_\lambda Y, \quad S_\lambda = X(X^\top X + \lambda I_p)^{-1} X^\top,$$

leave-one-out cross-validation (LOOCV) can be computed efficiently. Let $r_i(\lambda) = y_i - \hat{y}_i(\lambda)$ be the i-th residual and $s_{ii}(\lambda)$ the i-th diagonal of S_λ. For a general linear smoother $\hat{Y} = SY$, one can show

$$\hat{y}_i^{(-i)}(\lambda) = y_i - \frac{r_i(\lambda)}{1 - s_{ii}(\lambda)},$$

where $\hat{y}_i^{(-i)}(\lambda)$ is the fitted value for observation i using a model trained on all data except (x_i, y_i). Then the LOOCV score is

$$\mathrm{CV}(\lambda) = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{y}_i^{(-i)}(\lambda)\right)^2 = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{r_i(\lambda)}{1 - s_{ii}(\lambda)}\right)^2.$$

We choose λ to minimize $\mathrm{CV}(\lambda)$.

21.5.2 Ridge paths and regularization trajectories

Because

$$\hat{\beta}_\lambda = V \,\mathrm{diag}\left(\frac{\sigma_1^2}{\sigma_1^2 + \lambda}, \ldots, \frac{\sigma_p^2}{\sigma_p^2 + \lambda}\right) V^\top \hat{\beta}_{\mathrm{OLS}},$$

each coordinate of $\hat{\beta}_\lambda$ in the V-basis is a smooth, monotone function of λ:

$$\tilde{\beta}_{\lambda,j} = \frac{\sigma_j^2}{\sigma_j^2 + \lambda} \tilde{\beta}_{\mathrm{OLS},j}.$$

As λ increases from 0 to ∞, each factor $\sigma_j^2/(\sigma_j^2 + \lambda)$ decreases from 1 to 0, so

$$\hat{\beta}_\lambda \to 0 \quad \text{and} \quad \mathrm{df}(\lambda) \to 0.$$

Plotting $\hat{\beta}_\lambda$ versus $\log \lambda$ gives ridge *paths* or *regularization trajectories*, which visualize how coefficients shrink and how model complexity decreases as λ grows.